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Tortorella, Jr.

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(54) **LICENSE PLATE HOLDER**

(56) **References Cited**

(71) Applicant: **Frank Joseph Tortorella, Jr.**, Park Ridge, IL (US)
(72) Inventor: **Frank Joseph Tortorella, Jr.**, Park Ridge, IL (US)
(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Related U.S. Application Data

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(60) Provisional application No. 62/880,056, filed on Jul. 29, 2019.

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G09F 9/30 (2006.01)
G09F 7/18 (2006.01)
G09F 21/00 (2006.01)

(52) **U.S. Cl.**
CPC **B60R 13/105** (2013.01); **G09F 9/30** (2013.01); **G09F 2007/1895** (2013.01); **G09F 21/00** (2013.01)

(58) **Field of Classification Search**
CPC B60R 13/105; G09F 9/30; G09F 21/00; G09F 2007/1895; G09F 13/18; G09F 7/18; G09F 21/04

See application file for complete search history.

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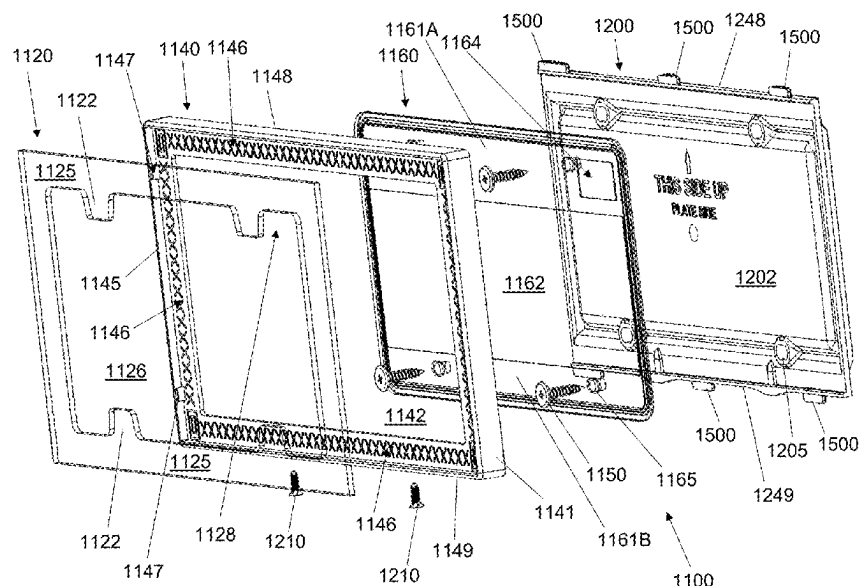
Primary Examiner — Cassandra Davis

(74) *Attorney, Agent, or Firm* — MARSHALL, GERSTEIN & BORUN LLP

(57) **ABSTRACT**

A license plate holder includes a frame defining a central opening to permit display of indicia on license plate display region(s) of a license plate and defining frame display region(s) adjacent the central opening. A transmissive substrate includes one or more connectors to removably connect the transmissive substrate to a license plate base. In an assembled state with the transmissive substrate removably connected to the license plate base via the connector(s) and the frame interposed between the transmissive substrate and the license plate base, a central region of the transmissive substrate is positioned in front of the central opening and the license plate display region(s) and a peripheral region of the transmissive substrate is positioned in front of the frame display region(s).

19 Claims, 24 Drawing Sheets



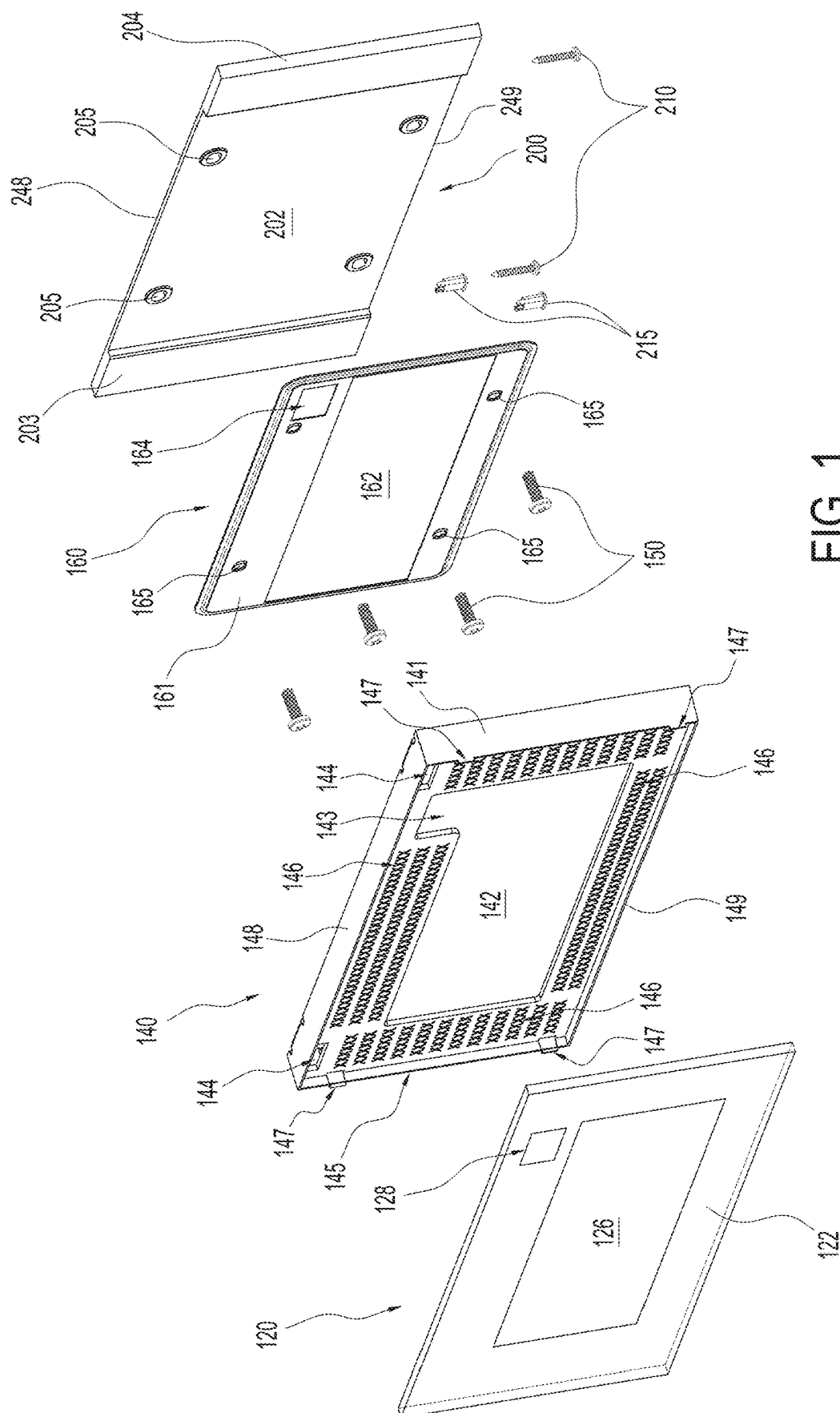


FIG. 1

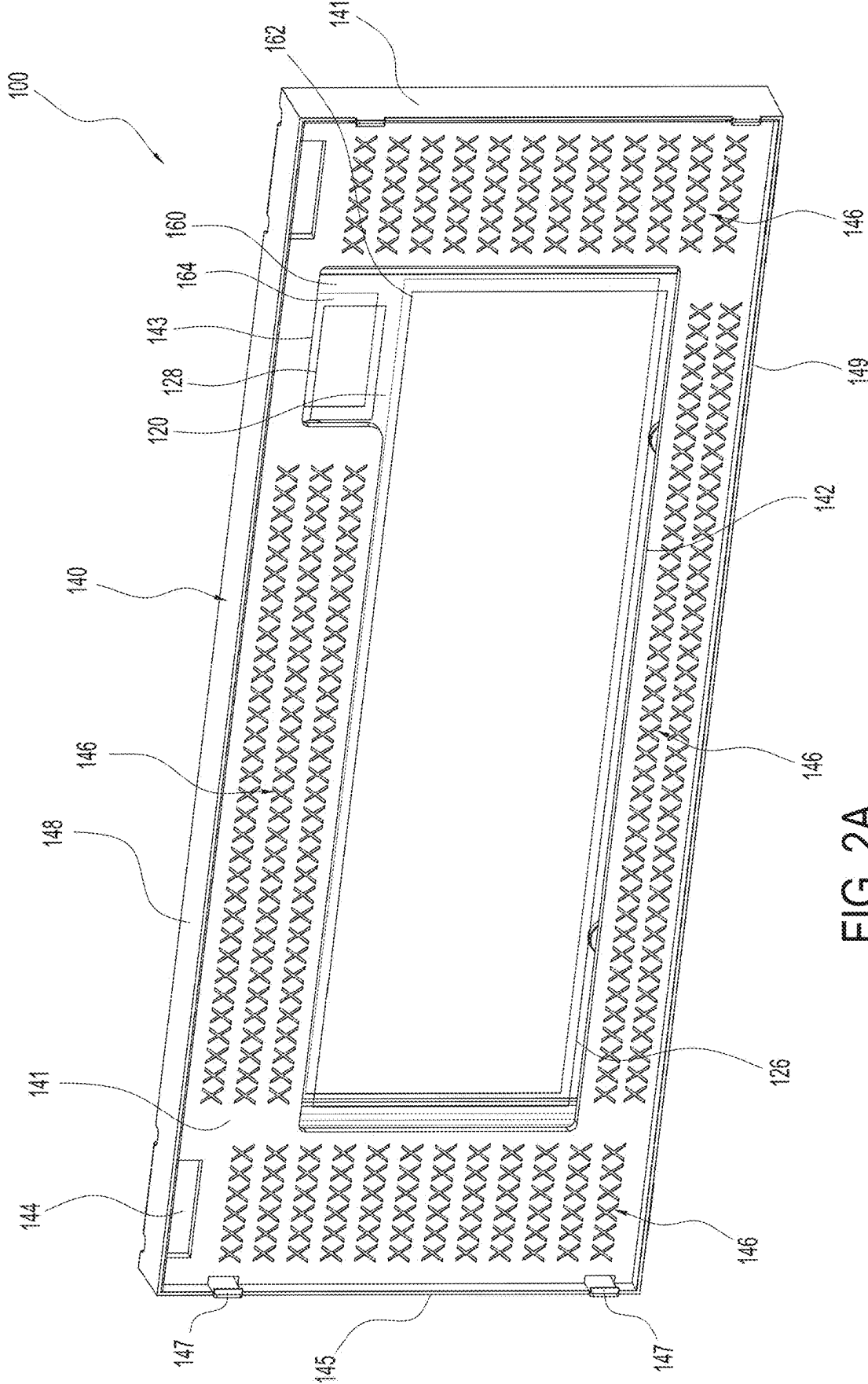


FIG. 2A

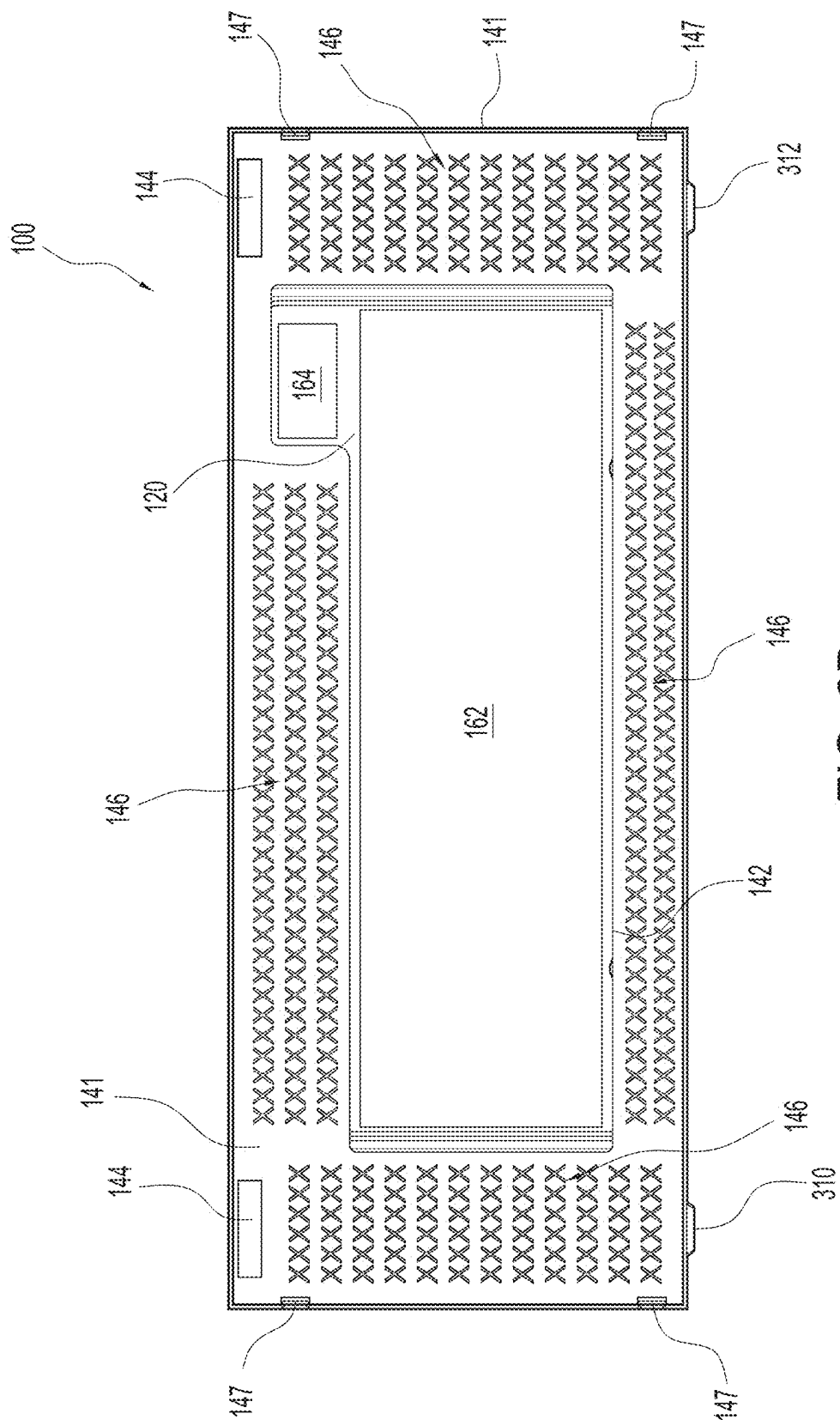


FIG. 2B

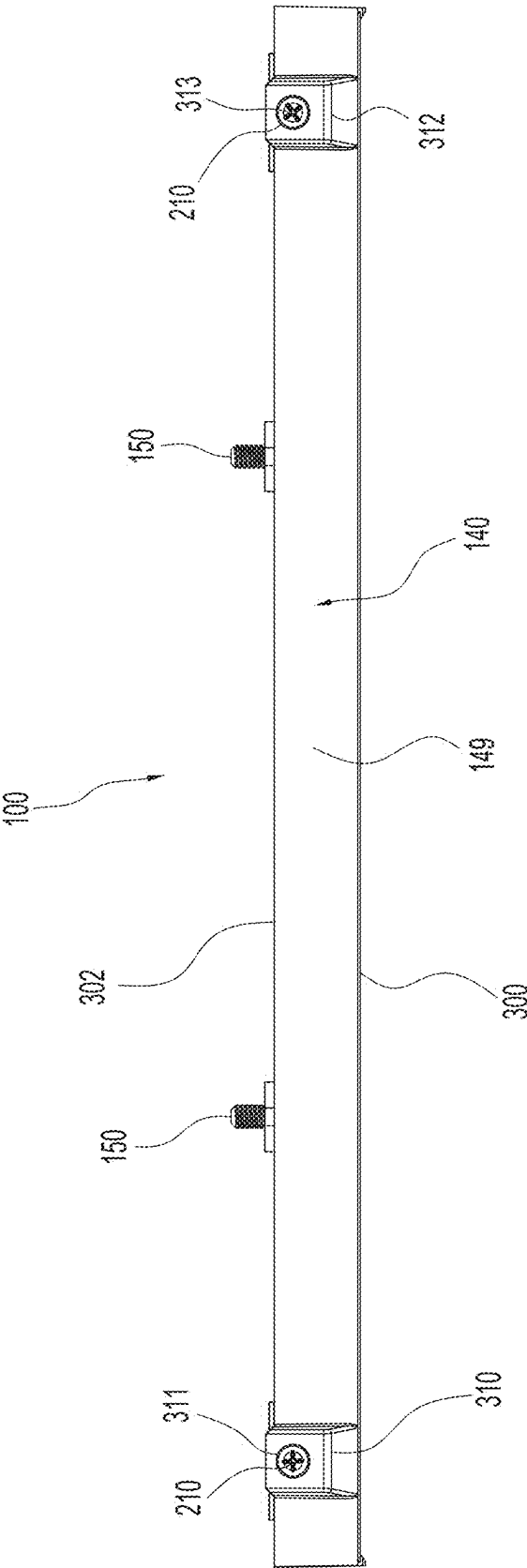


FIG. 3

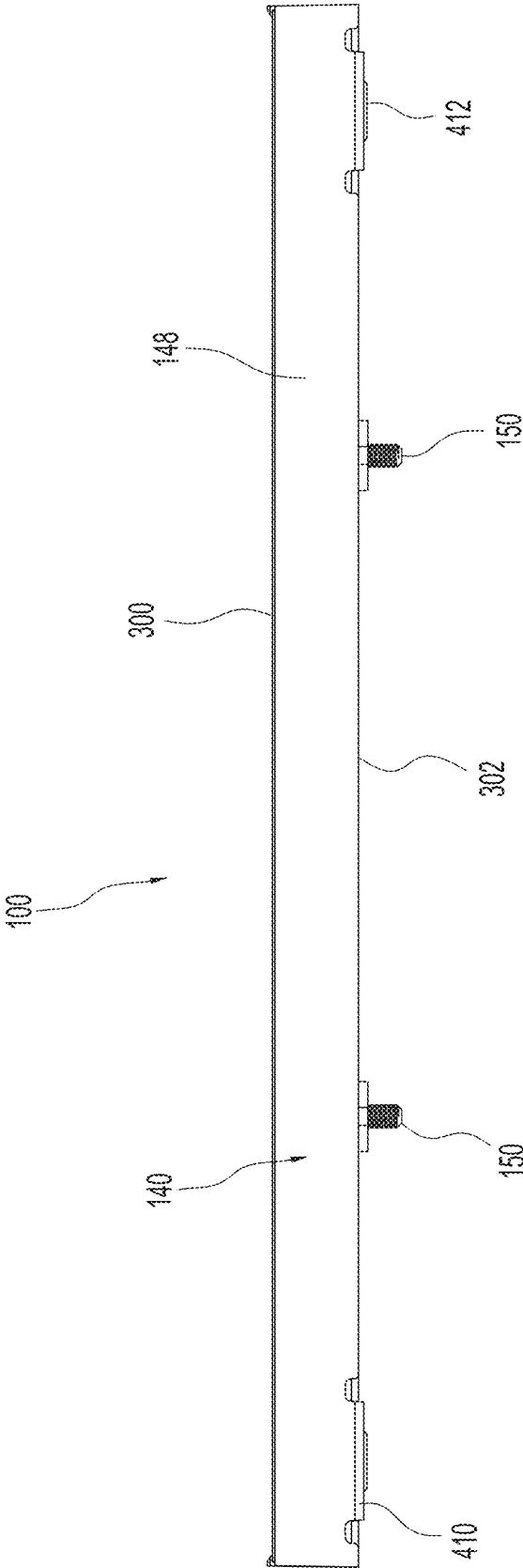
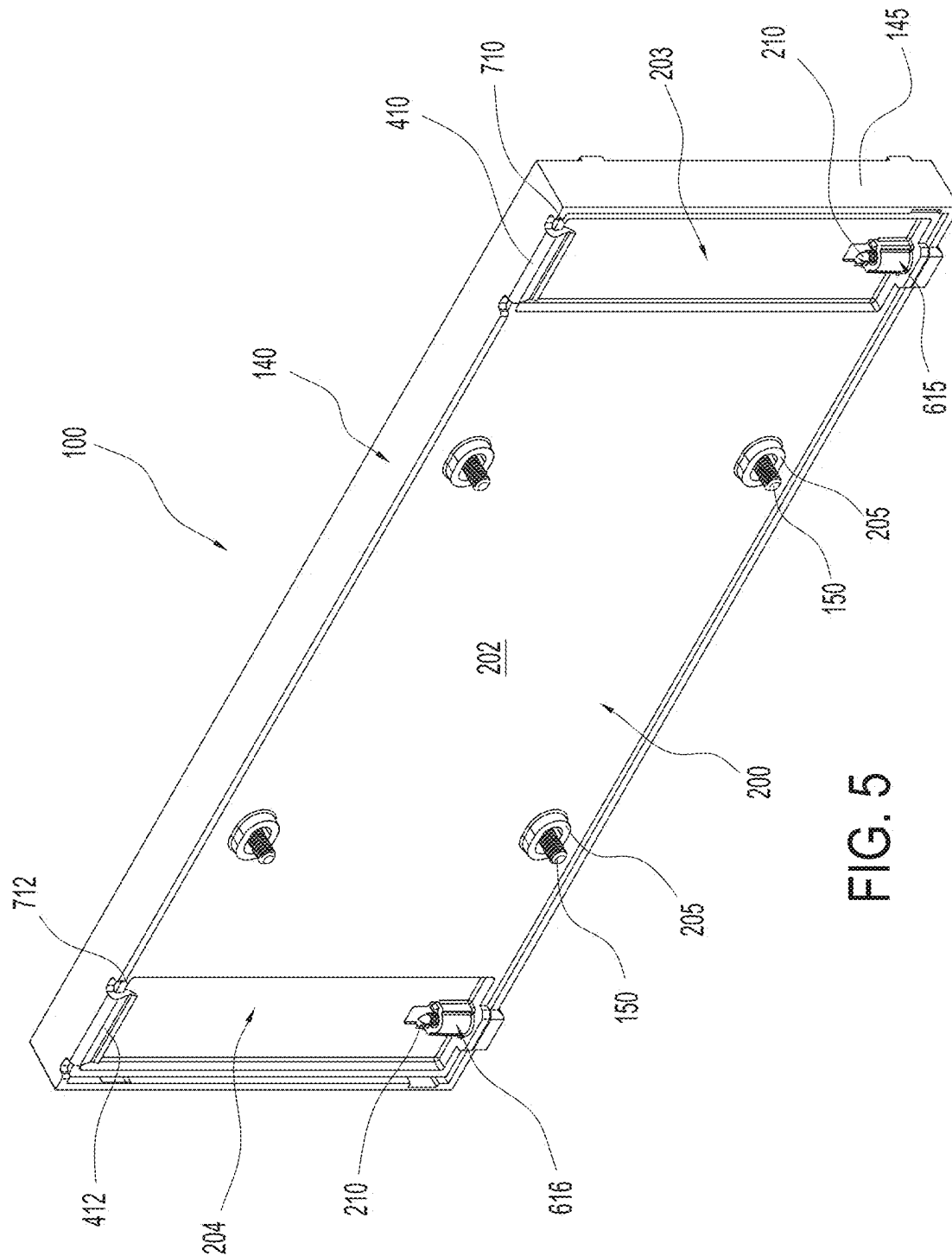


FIG. 4



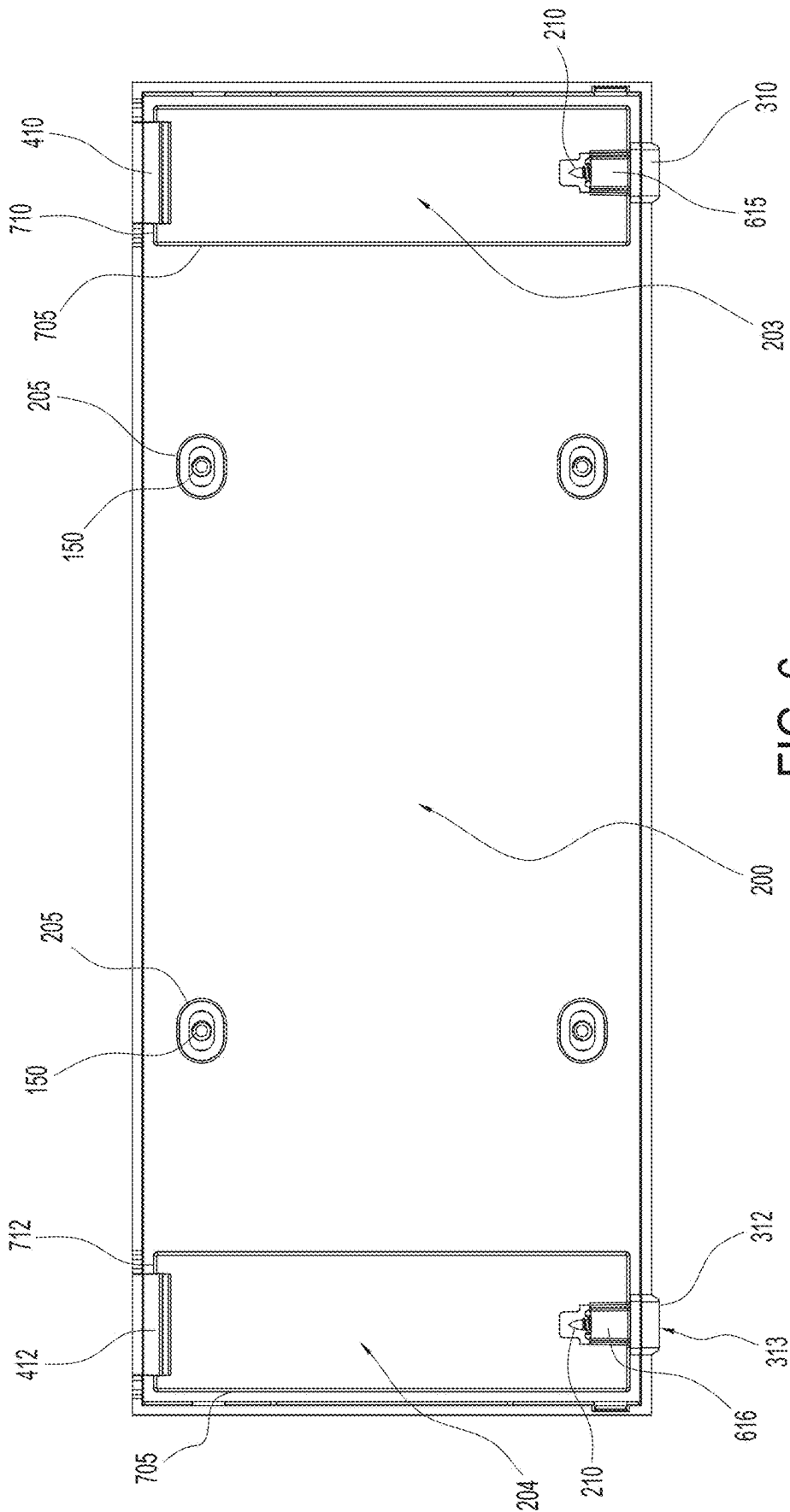


FIG. 6

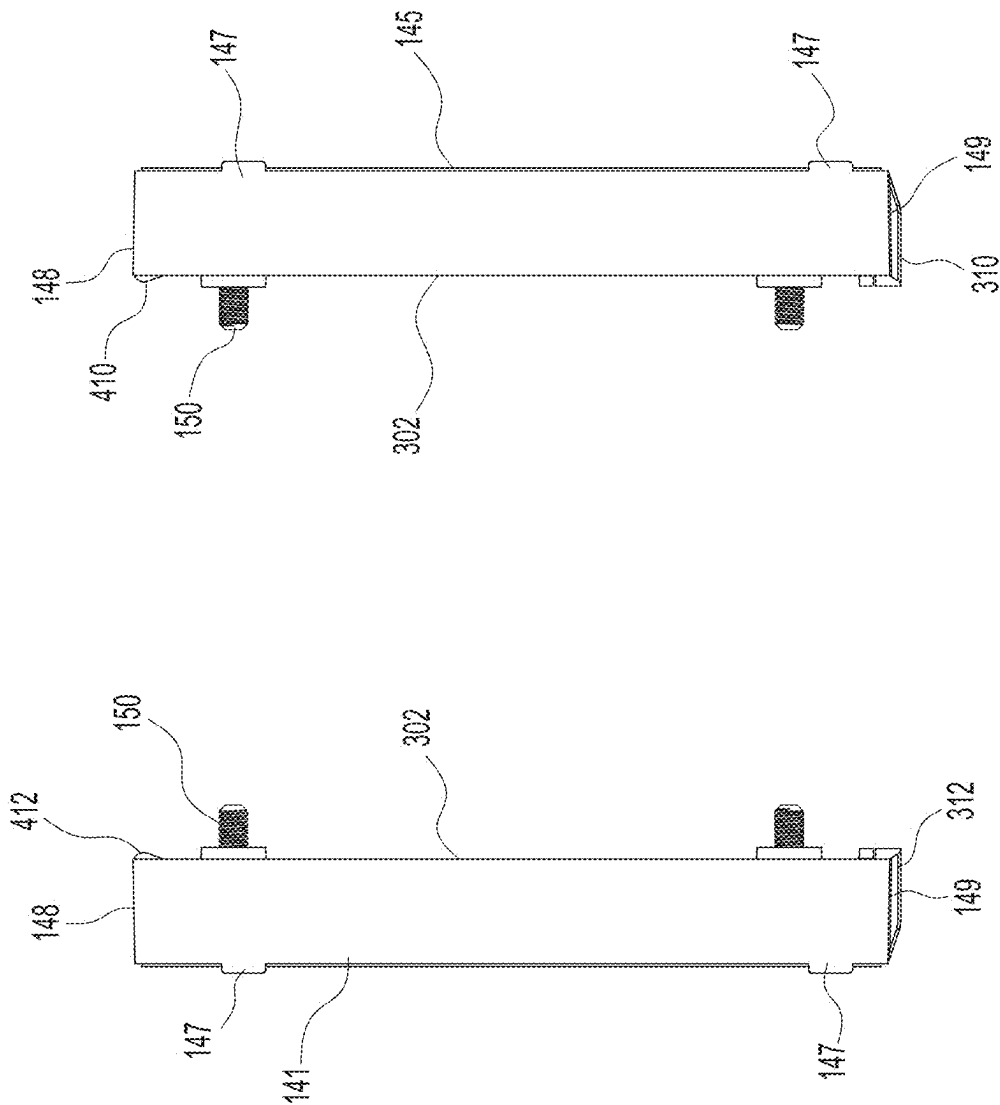
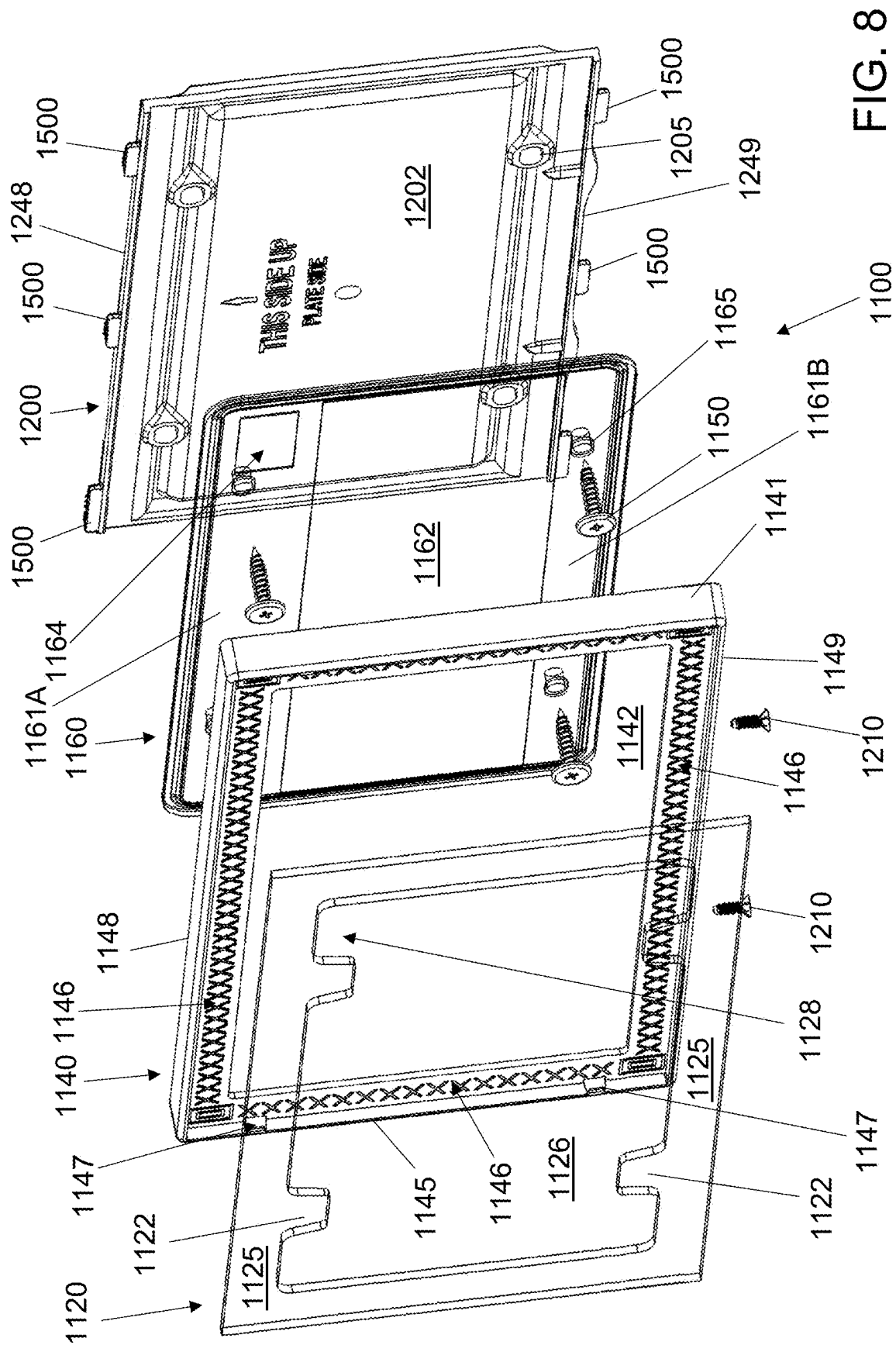


FIG. 7B

FIG. 7A



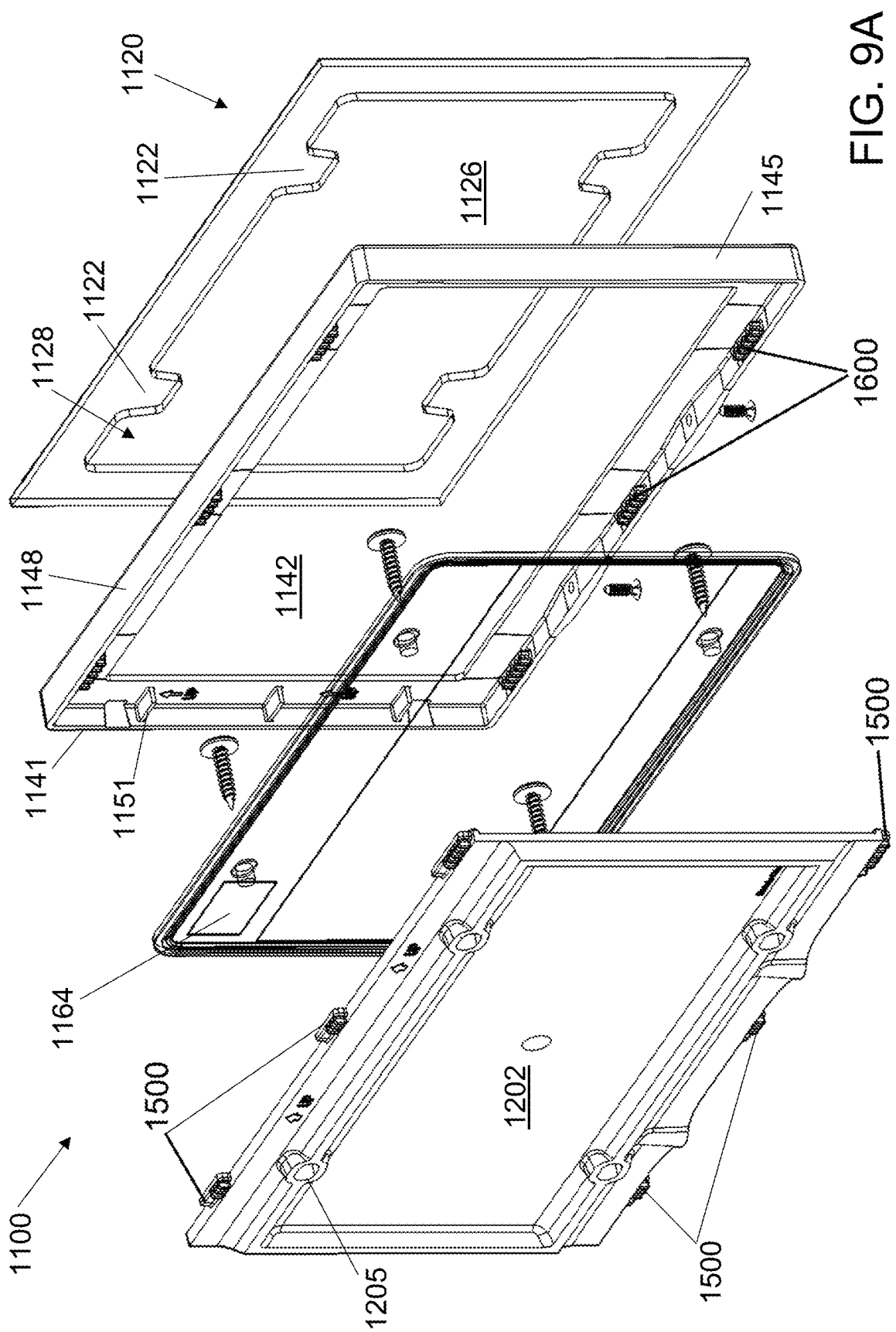


FIG. 9A

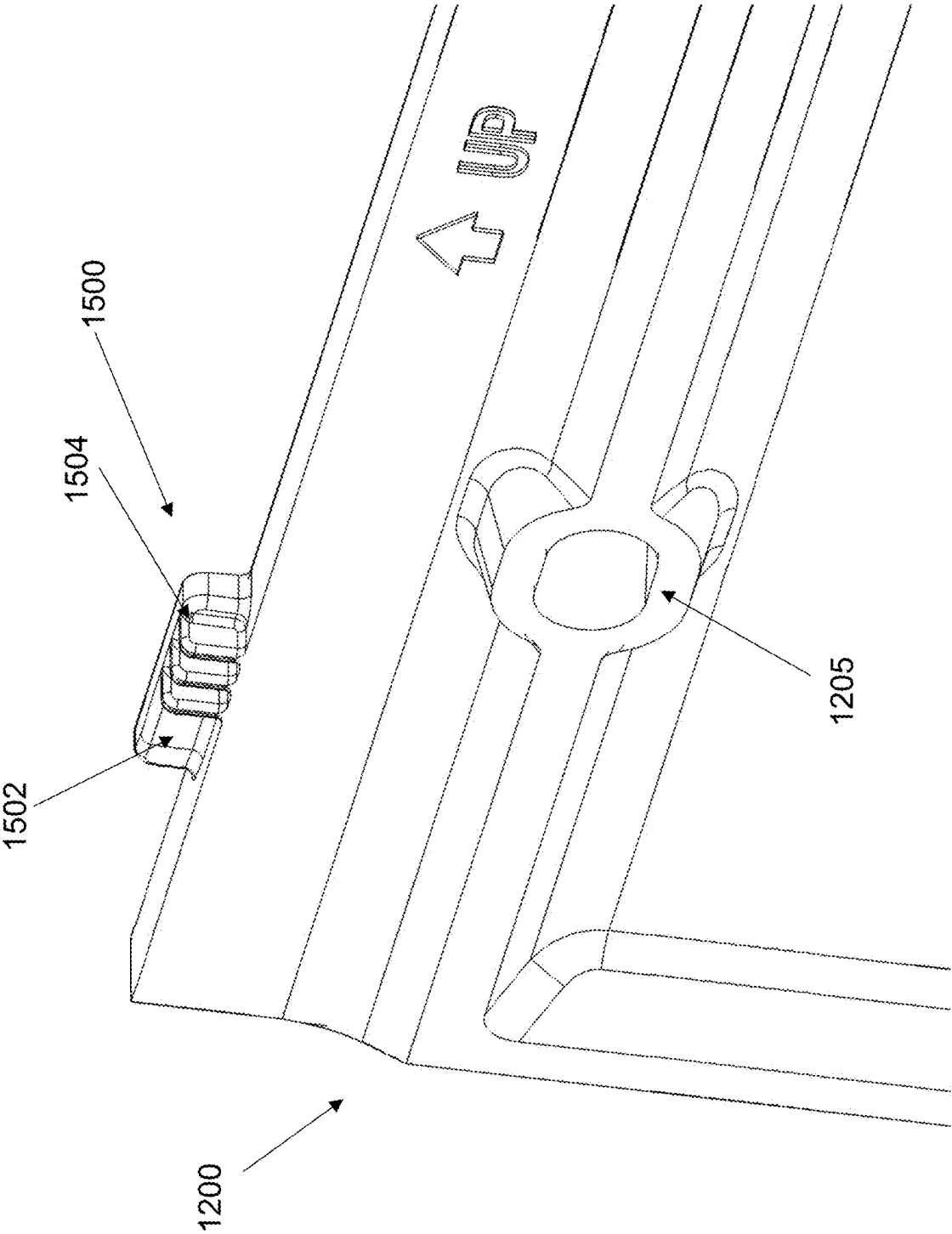


FIG. 9B

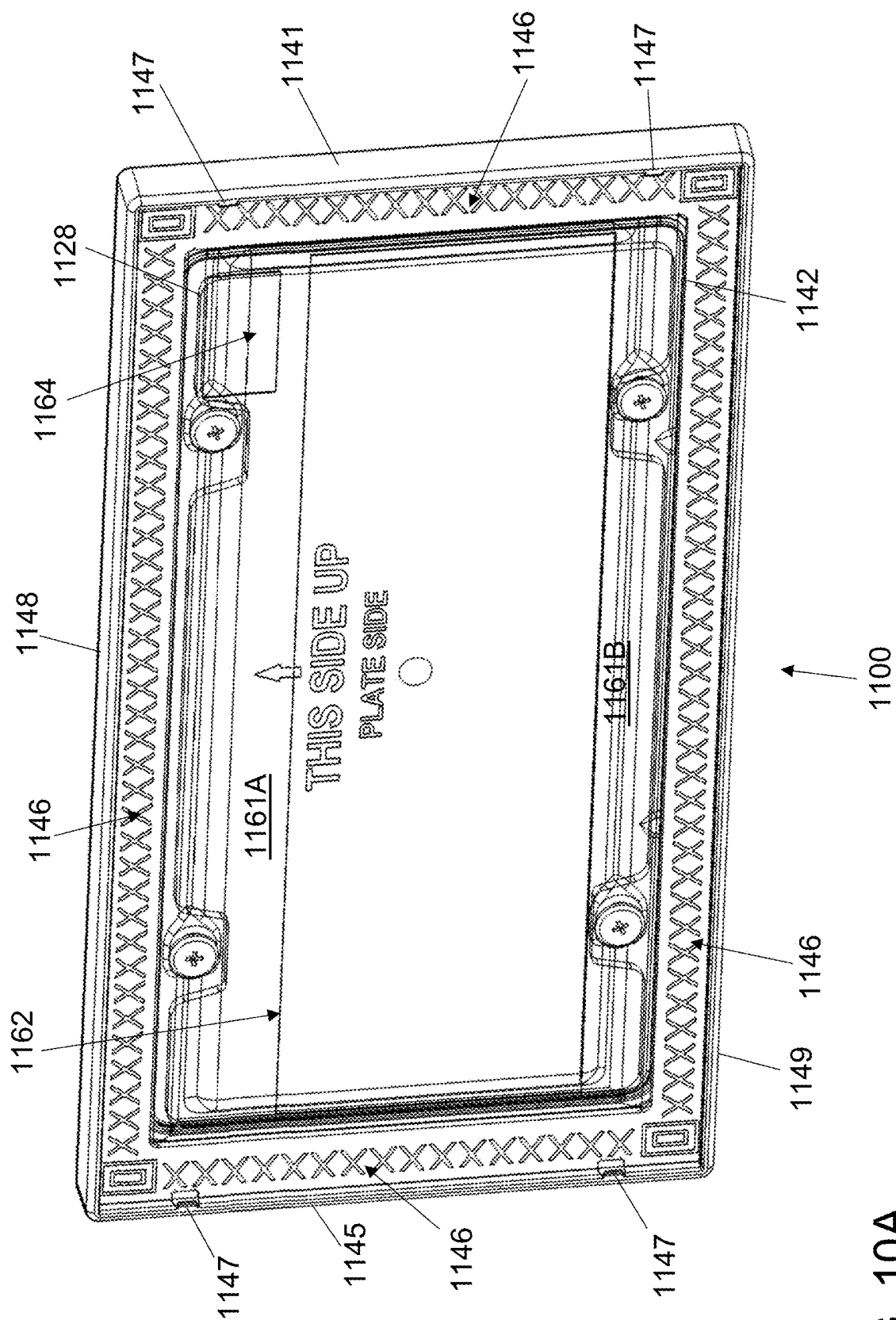


FIG. 10A

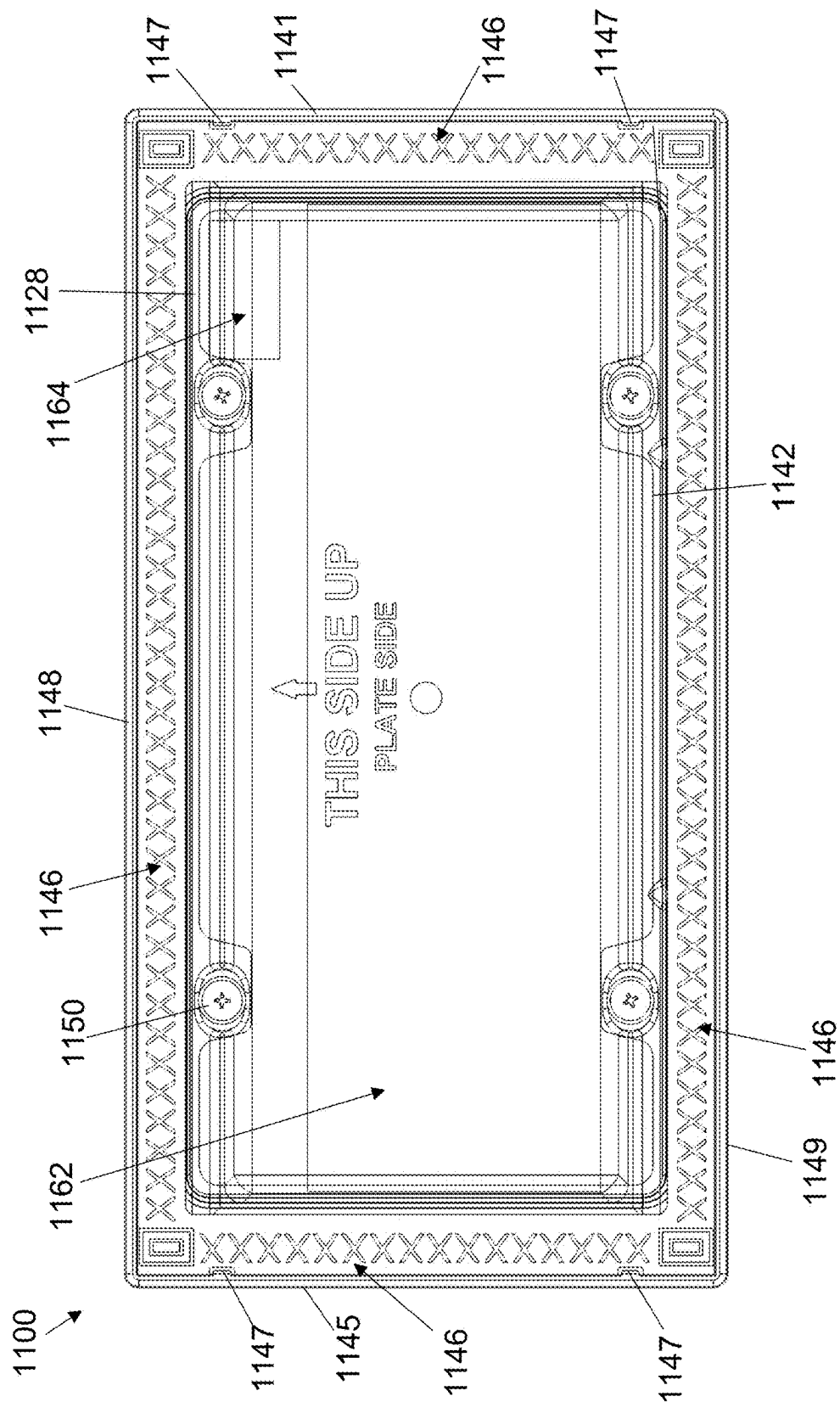


FIG. 10B

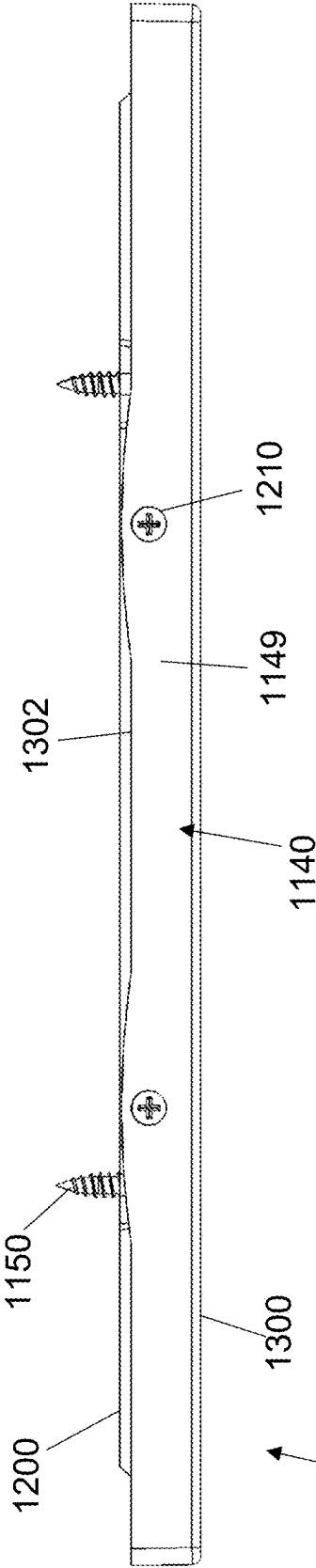


FIG. 11

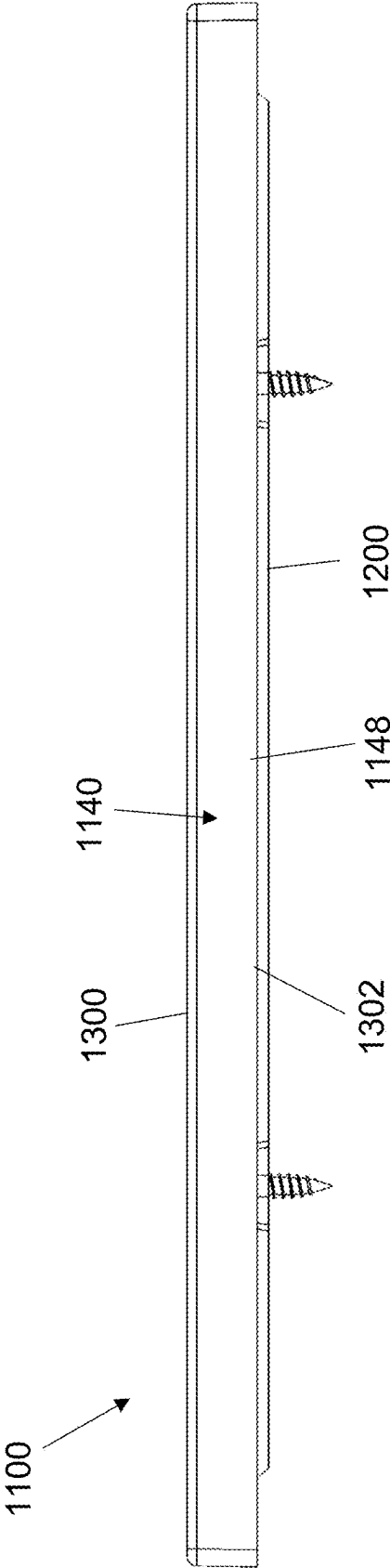


FIG. 12

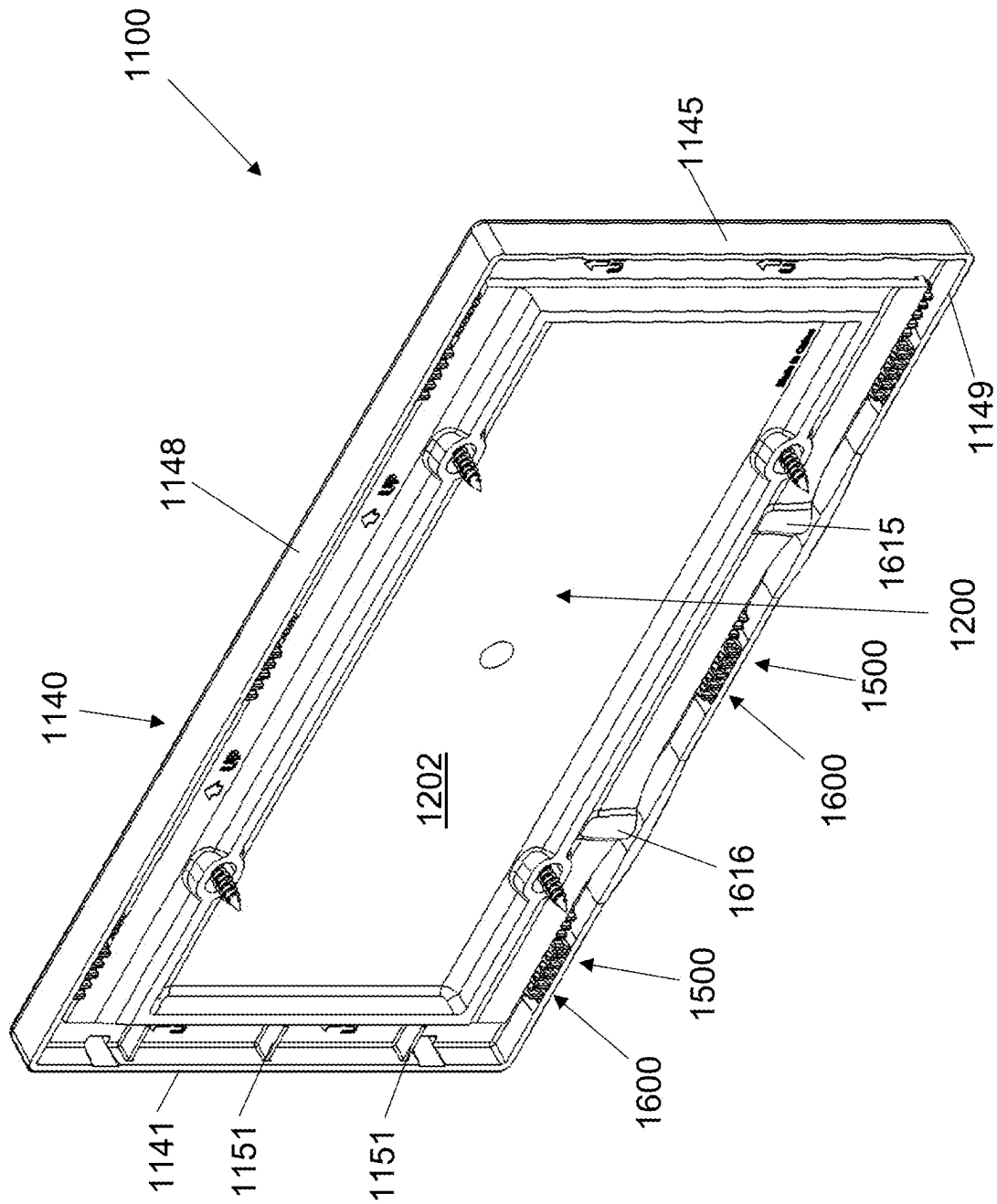
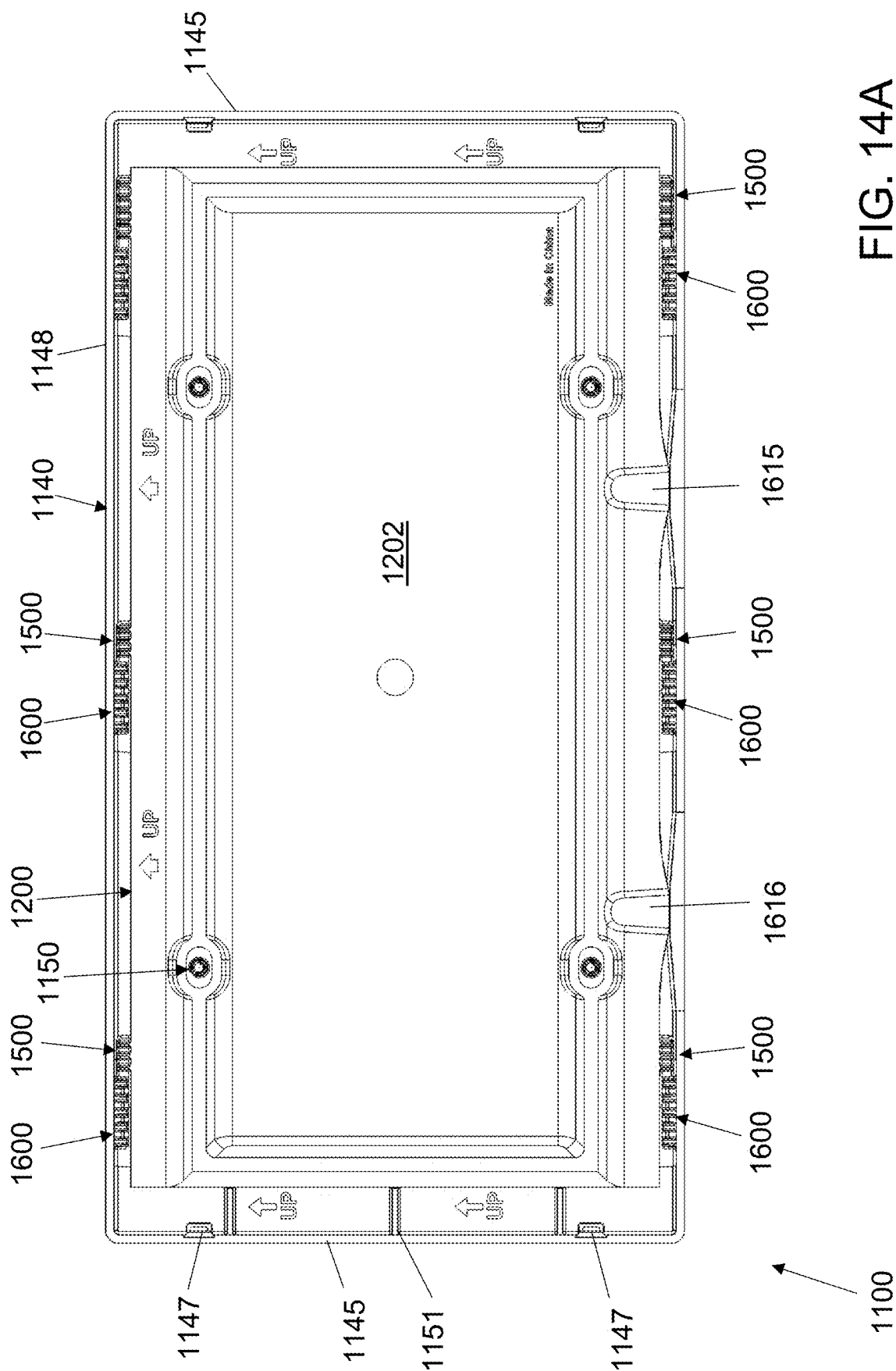
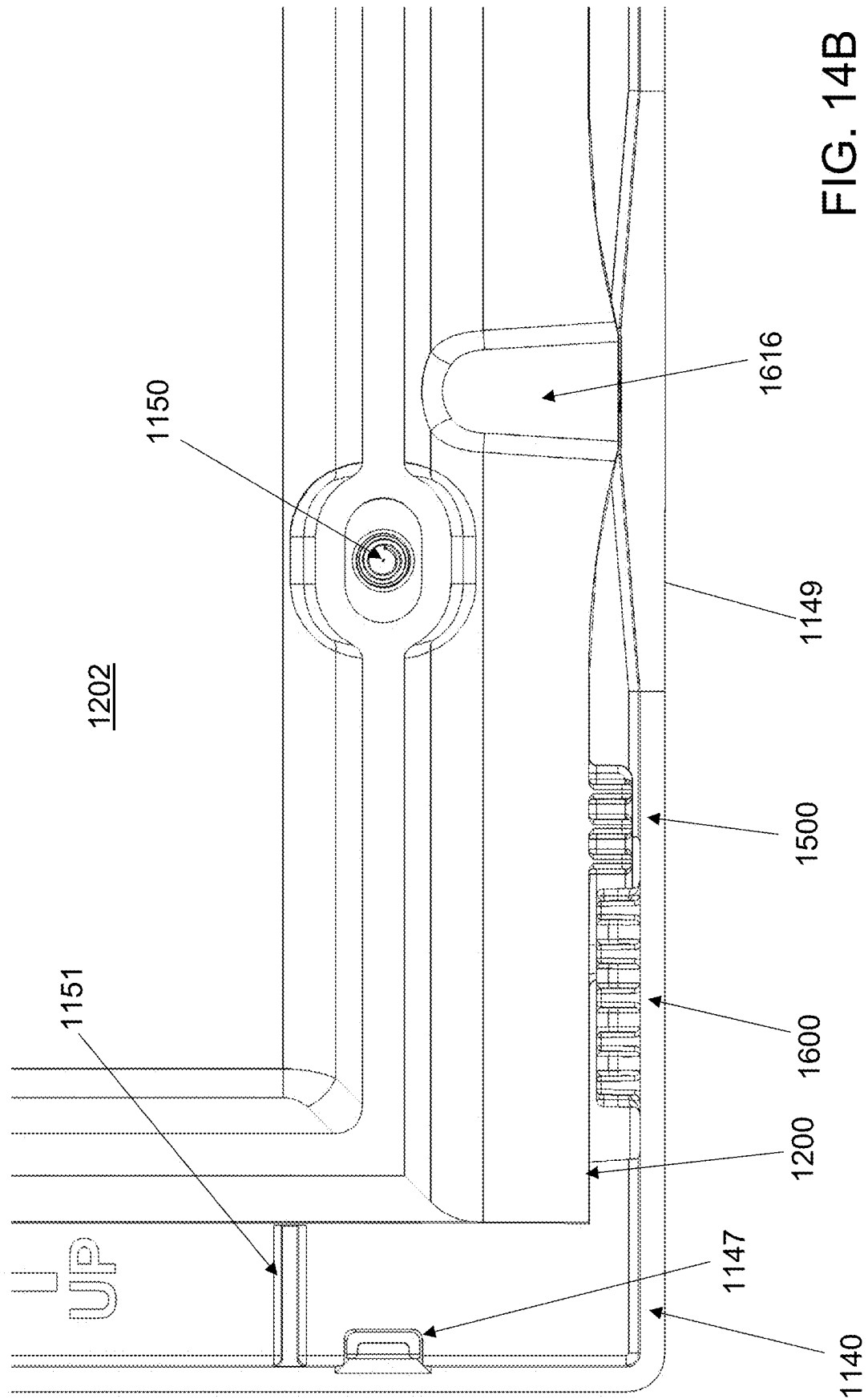
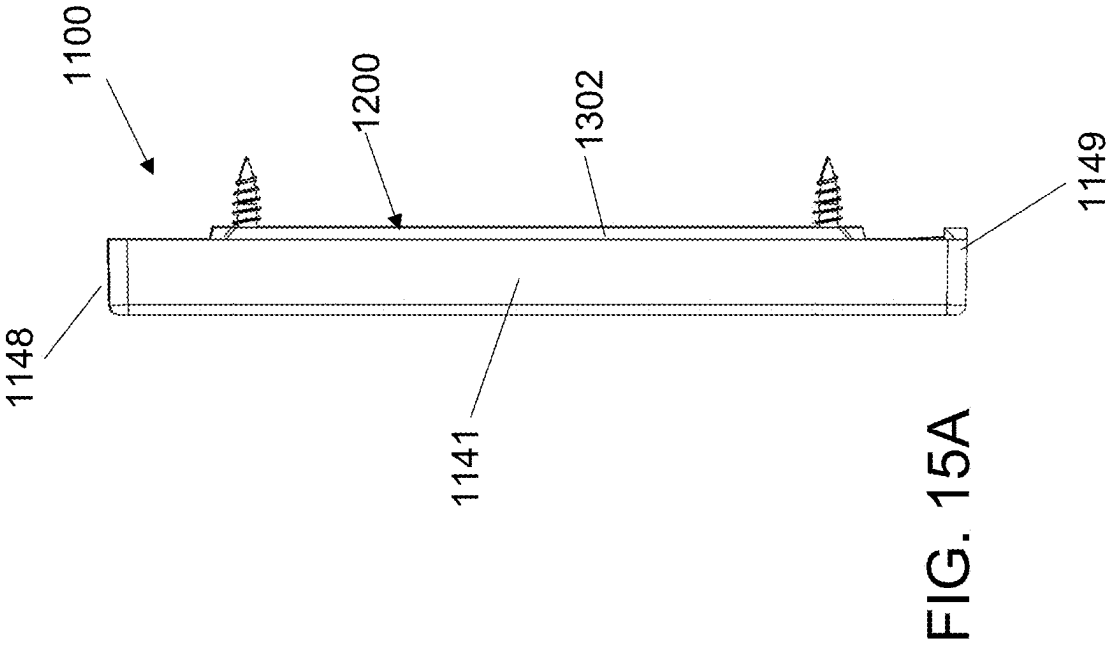
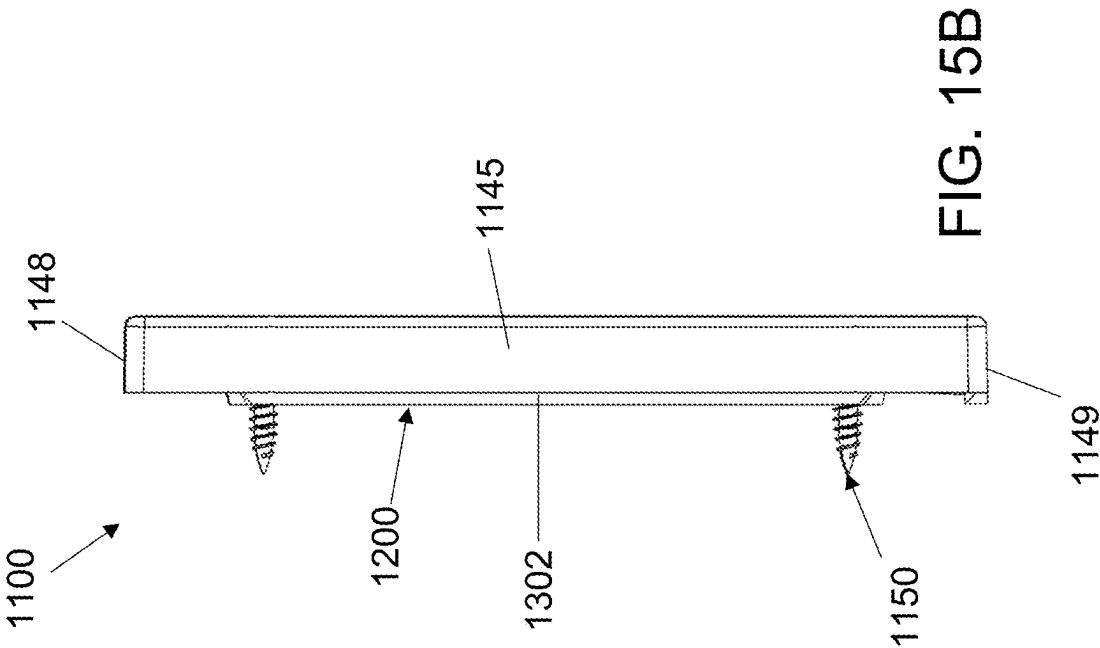


FIG. 13







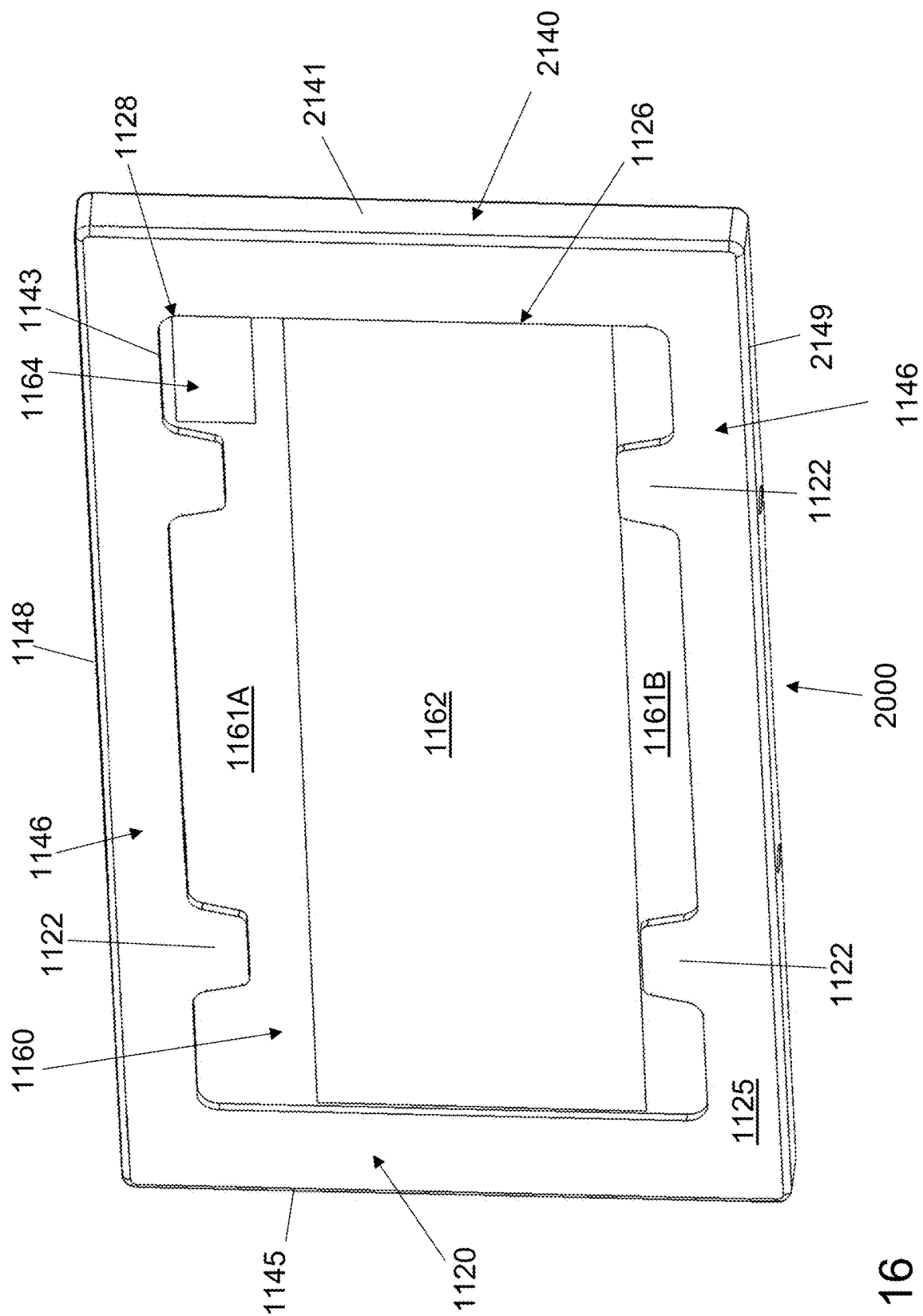


FIG. 16

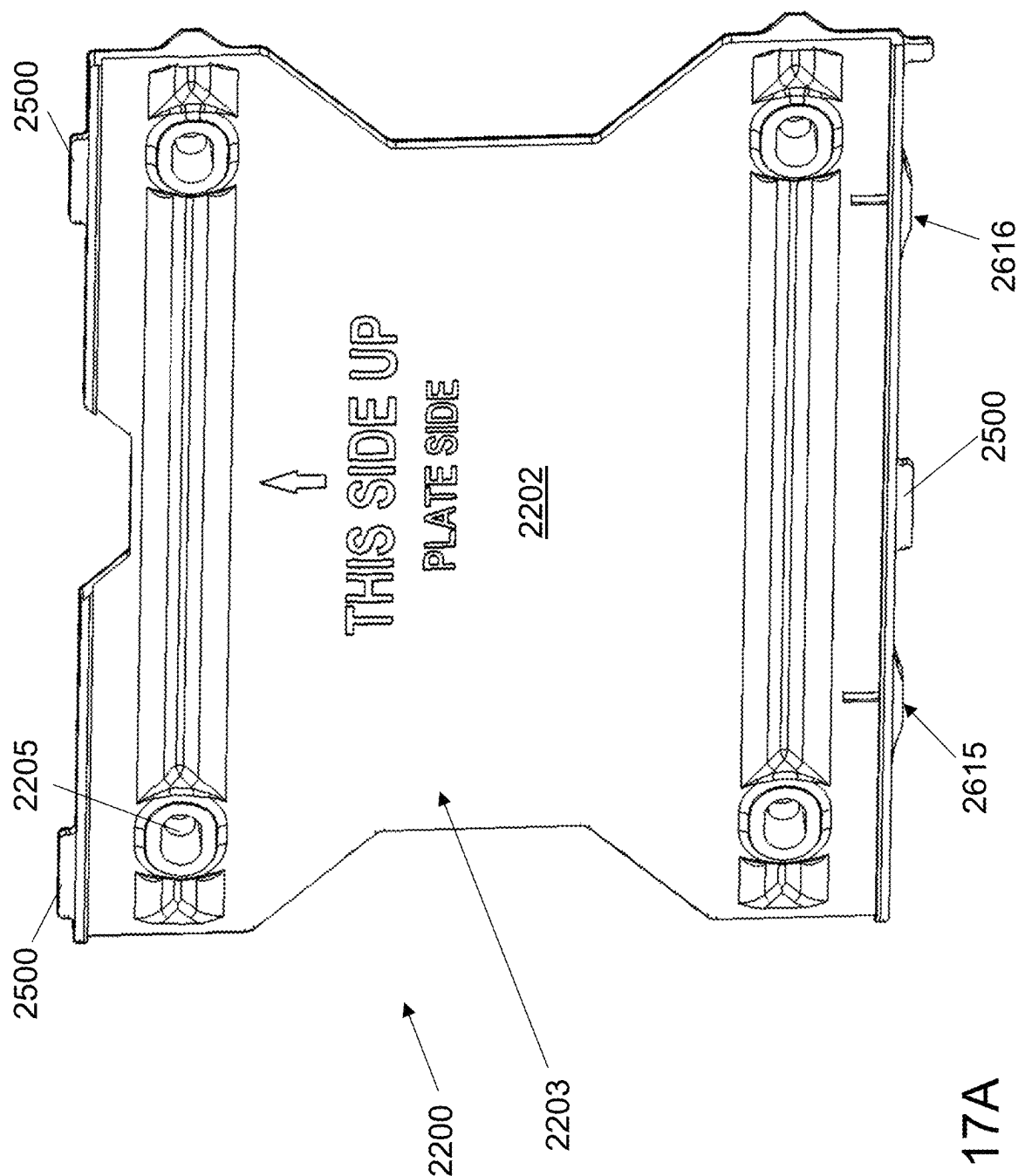


FIG. 17A

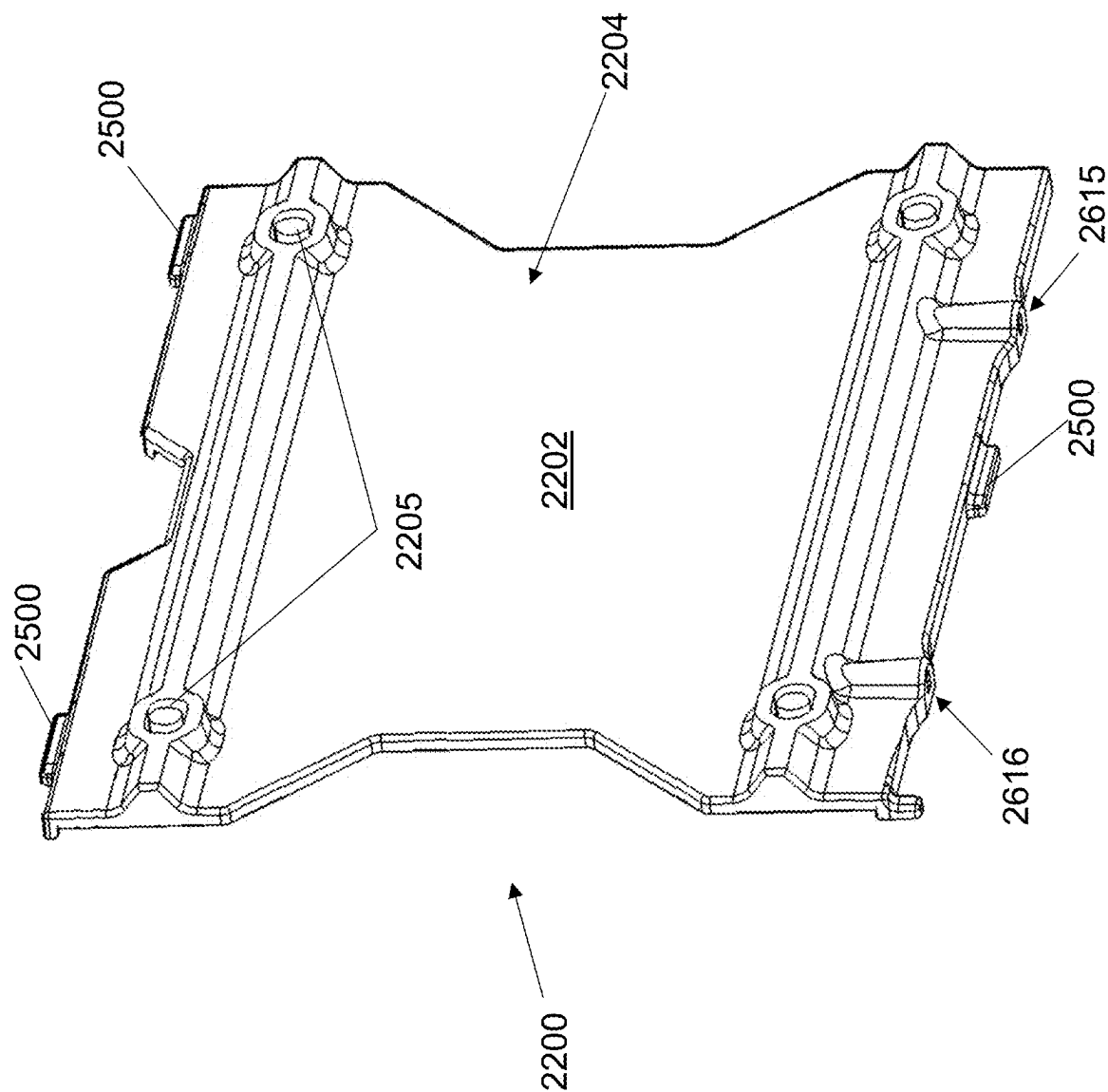


FIG. 17B

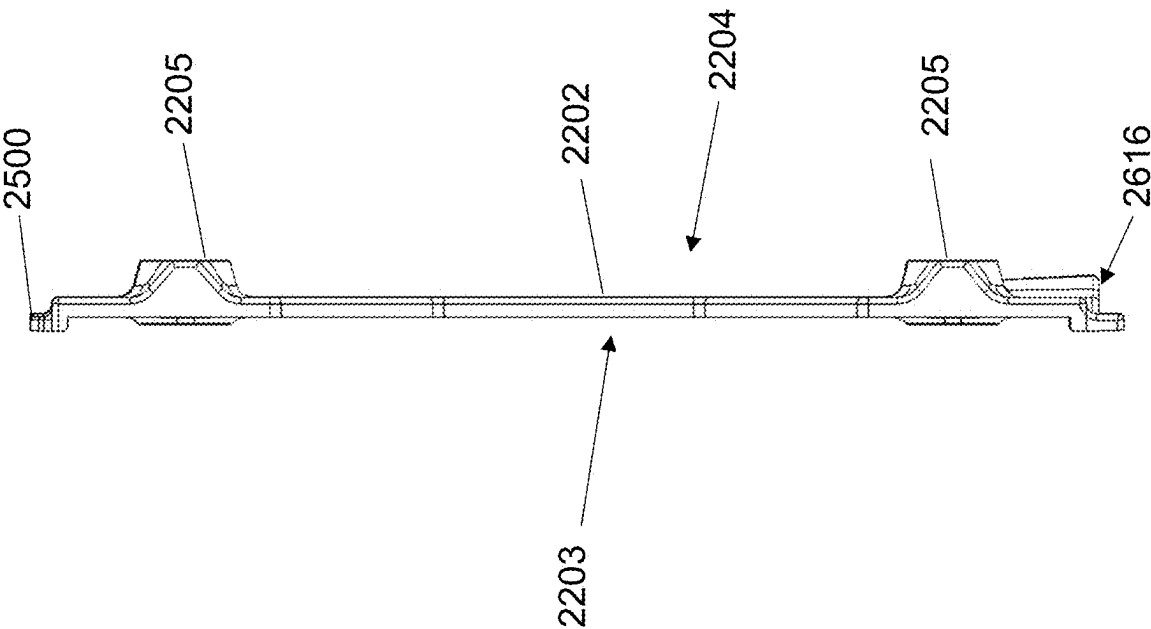


FIG. 17C

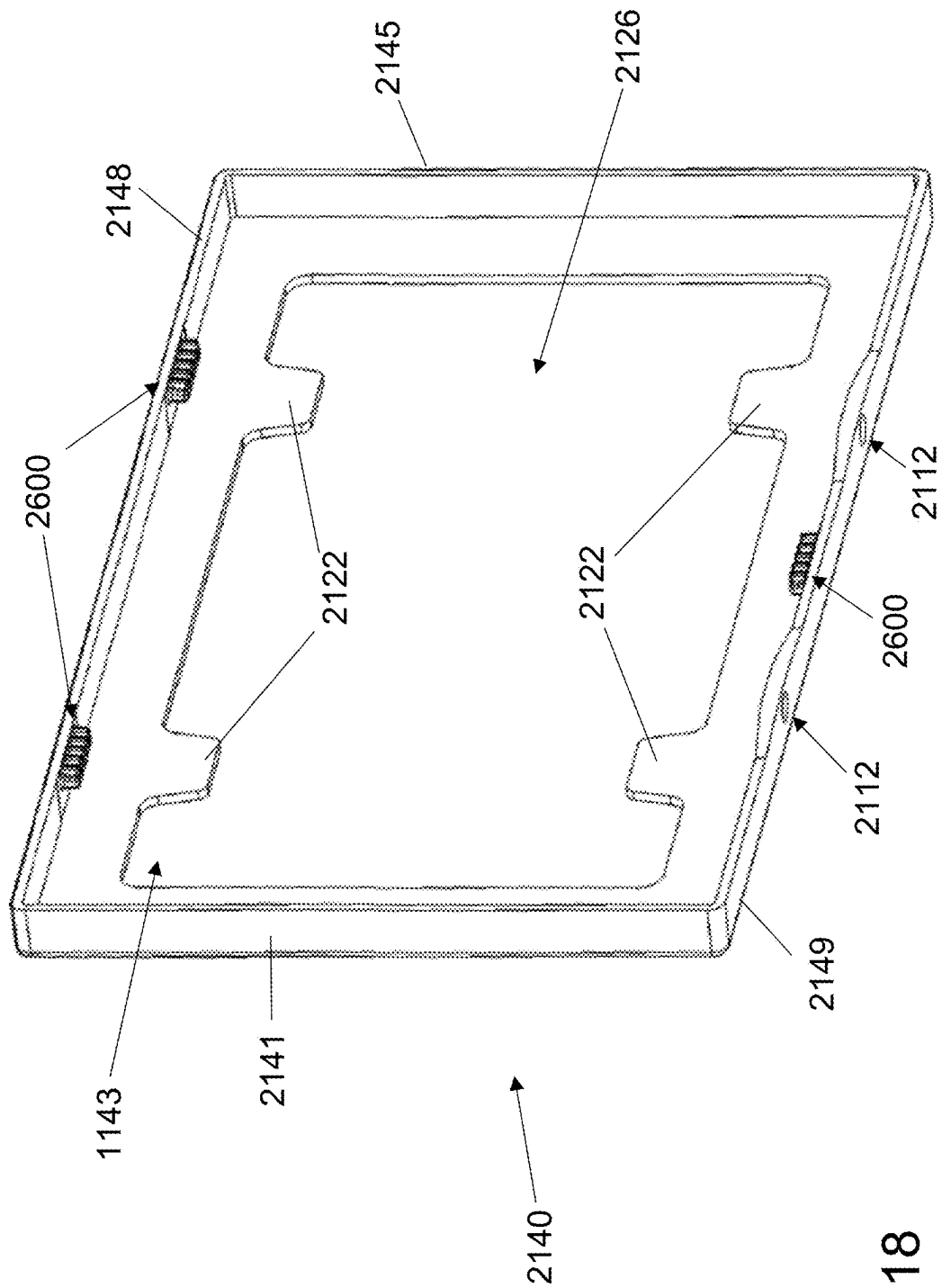


FIG. 18

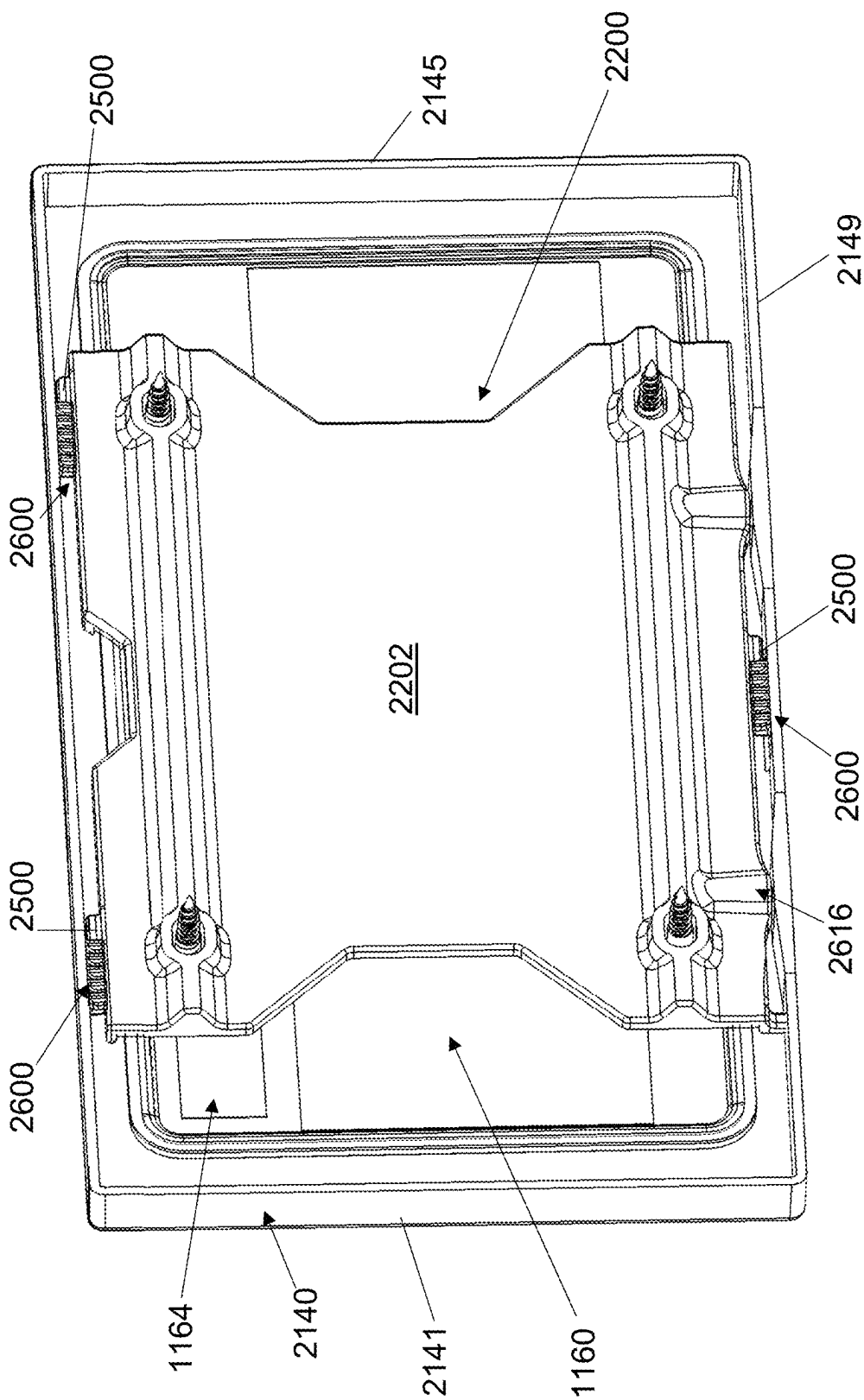


FIG. 19

LICENSE PLATE HOLDER

CROSS-REFERENCE

This application claims priority to, and is a continuation of U.S. patent application Ser. No. 17/630,945, which will issue as U.S. Pat. No. 12,030,438 on Jul. 9, 2024, a national phase application under 35 U.S.C. 371 to PCT Appl. Ser. No. PCT/US2020/043185, filed on Jul. 23, 2020, which claimed priority to U.S. Provisional Pat. Appl. Ser. No. 62/880,056, filed on Jul. 29, 2019, the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND

A vehicle registration plate, also known as a number plate (British) or a license plate (American), is a metal or plastic plate attached to a motor vehicle or trailer for official identification purposes. All countries require registration plates for road vehicles such as cars, trucks, and motorcycles. The registration identifier is a numeric or alphanumeric ID that uniquely identifies the vehicle owner within the issuing region's vehicle register. In various examples, license plates may be 305 mm×152 mm (12.0"×6.0") (the United States), 520 mm×110 mm (20.5"×4.3") (various European countries), or 440 mm×140 mm (17.3"×5.5") (China). License plates may be formed from a variety of materials, such as plastic or metal (e.g., aluminum, steel, an alloy, etc.).

To secure the license plate to the vehicle, a license plate holder or license plate frame is used to hold the license plate and is itself attached to the vehicle, typically by screws that mount into threaded fittings on the vehicle. The license plate holder is configured to ensure that the license plate numbers and regulatory stickers are not obscured. License plate holders are often made from a metal (e.g., stainless steel, aluminum, etc.) or plastic, and may have a protective coating applied thereover.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an exploded perspective view of an example license plate holder in accord with at least some of the present concepts.

FIG. 2A is a perspective view of the example license plate holder of FIG. 1 in an assembled state.

FIG. 2B is a front view of the example license plate holder of FIG. 1 in an assembled state.

FIG. 3 is an example bottom view of the example license plate holder of FIGS. 1-2B in an assembled state.

FIG. 4 is an example top view of the example license plate holder of FIGS. 1-3 in an assembled state.

FIG. 5 is an example rear perspective view of the example license plate holder of FIGS. 1-4 in an assembled state.

FIG. 6 is an example rear view of the example license plate holder of FIGS. 1-5 in an assembled state.

FIGS. 7A-7B are, respectively, example right side and left side views of the example license plate holder of FIGS. 1-6 in an assembled state.

FIG. 8 is an example exploded front perspective view of a second embodiment of an example license plate holder in accord with at least some of the present concepts.

FIG. 9A is an example rear exploded perspective view of the example license plate holder of FIG. 8.

FIG. 9B is an example rear close-up view of a portion of the example license plate holder of FIG. 9A showing an example connector.

FIG. 10A is a front perspective view of the example license plate holder of FIGS. 8-9B in an assembled state.

FIG. 10B is a front view of the example license plate holder of FIGS. 8-10A in an assembled state.

FIG. 11 is an example bottom view of the example license plate holder of FIGS. 8-10B in an assembled state.

FIG. 12 is an example top view of the example license plate holder of FIGS. 8-11 in an assembled state.

FIG. 13 is an example rear perspective view of the example license plate holder of FIGS. 8-12 in an assembled state.

FIG. 14A is an example rear view of the example license plate holder of FIGS. 8-13 in an assembled state.

FIG. 14B is a close-up rear view of a portion of the example license plate holder of FIG. 14A showing an example connection between components of the example license plate holder.

FIGS. 15A-15B are, respectively, example right side and left side views of the example license plate holder of FIGS. 8-14B in an assembled state.

FIG. 16 is an example front perspective view of an example third embodiment of an example license plate holder in an assembled state in accord with at least some of the present concepts.

FIG. 17A is an example front perspective view of an example base usable in the example license plate holder of FIG. 16.

FIG. 17B is an example rear perspective view of the example base of FIG. 17A.

FIG. 17C is an example side view of the example base of FIGS. 17A-17B.

FIG. 18 is a rear perspective view of an example frame for the example license plate holder of FIGS. 16-17C.

FIG. 19 is a rear perspective view of the example license plate holder of FIGS. 16-18 in an assembled state.

The figures are not necessarily to scale. Wherever possible, the same reference numbers, or similar reference numbers, are used throughout the drawing(s) and accompanying written description to refer to the same or like parts. As used herein, stating that any part is in any way positioned on (e.g., positioned on, located on, disposed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween. Stating that any part is in contact with another part means that there is no intermediate part between the two parts.

DETAILED DESCRIPTION

In some aspects, this disclosure relates to an example license plate holder for vehicles, inclusive of road vehicles (e.g., cars, trucks, motorcycles, scooters, etc.), off-road vehicles (e.g., 3-wheelers, all-terrain vehicles, four wheelers, etc.), and marine vehicles (e.g., boats), which require registration with a governmental body and display of a license plate on the vehicle.

FIG. 1 is an exploded perspective view of an example license plate holder 100 in accord with at least some of the present concepts. The example license plate holder 100 is shown to include, from left-to-right, an example transmissive substrate 120, an example frame 140 and an example base 200. An optional example license plate 160 is also shown in relation to the example license plate holder 100 for contextual illustration to show an example integration of a license plate 160 with a license plate holder 100 in accord with at least some aspects of the present concepts.

The example transmissive substrate **120** may include, for instance, but is not limited to, a substrate of glass, treated or strengthened glass (e.g., Gorilla Glass (Corning), Dragontrail (AGC Inc.), Xensation (Schott AG)), Plexiglass (e.g., Lucite, Acrylite, Perspex, Oroglass Optix, Altuglass, etc.), polycarbonate, acrylic (e.g., injection molded acrylic or polymethyl methacrylate, polystyrene or poly(methyl methacrylate) (PMMA)). In some examples, the example transmissive substrate **120** is a uniform substrate of a selected material. In some examples, the example transmissive substrate **120** includes a plurality of different regions such as an example first region **126** (e.g., a central region) having a first set of characteristics (e.g., optical properties such as refractive index, transmission/transmittance, absorption, graphics, mechanical properties such as Young's modulus, surface finish, strength, hardness, impact resistance, chemical resistance, etc.), an example second region **122** (e.g., a peripheral region) having a second set of characteristics, and an example third region **128** having a third set of characteristics. In some examples, two regions may have the same characteristics (e.g., the example first region **126** and the example third region **128**). In the example transmissive substrate **120** shown in FIG. 1, the example first region **126** generally corresponds in size to a display area (e.g., a DIN 1451 license plate number) of an example license plate (e.g., example license plate **160**) to be incorporated into the example license plate holder **100** (e.g., a United States license plate, a European license plate, a Chinese license plate, etc.). For instance, in some examples, the example first region **126** is less than 12.0"×6.0" (305 mm×152 mm) for a United States license plate, less than 20.5"×4.3" (520 mm×110 mm) for a license plate of some European countries, or 17.3"×5.5" (440 mm×140 mm) for a Chinese license plate. In some examples, the example first region **126** is a cutout in the example transmissive substrate **120**.

In some examples, the example transmissive substrate **120** comprises a substrate of glass that is about 0.12 inches thick. In other examples, the example transmissive substrate **120** comprises a substrate of glass that is less than 0.12 inches thick or greater than 0.12 inches thick, such as between 0.12-0.25 inches thick.

In some examples, the third example third region **128** is optionally omitted. In some examples, the example transmissive substrate **120** includes the third example third region **128**, as may be required in jurisdictions requiring the display of an additional emblem or sticker (e.g., a governmental vehicle registration sticker, etc.). The example third region **128** may alternatively be sized differently (e.g., larger, smaller, etc.) and/or disposed in another location on the example transmissive substrate **120** to comport with sizing and/or location requirements of any particular jurisdiction. For instance, the example third region **128** may be disposed in a location other than shown in the example of FIG. 1, such as in an upper left portion of the example transmissive substrate **120** or in an upper center portion of the example transmissive substrate **120**. In some examples, the example third region **128** is a cutout in the example transmissive substrate **120**. In some examples, two example third regions **128** may be provided, such as to permit the display of two emblems or stickers.

The example frame **140** defines an example central opening **142** to permit display of indicia, such as the license plate number, on an example first license plate display region **162**. In some examples, such as is shown in FIG. 1, the example central opening **142** opens into, or is contiguous to, an example secondary opening **143** adjacent the example central opening **142**. In some examples, the example central

opening **142** is a distinct opening separate from that of the example secondary opening **143**. As noted above, in some examples, the third example third region **128** is optionally omitted where there are no jurisdictional requirements for the display of an additional emblem or sticker (e.g., a governmental vehicle registration sticker, etc.), at an example second license plate display region **164**, in an offset position from the example first license plate display region **162**. In such examples, the secondary opening **143** may likewise be omitted. In other examples, the example secondary opening **143** may alternatively be sized differently (e.g., larger, smaller, etc.) and/or disposed in another location on the example frame **140** to comport with sizing and/or location requirements of any particular jurisdiction. For instance, the example secondary opening **143** may be disposed in a location other than shown in the example of FIG. 1, such as in an upper left portion of the example frame **140** or in an upper center portion of the example frame **140**.

In some examples, the example central opening **142** is less than about 305 mm×152 mm (12.0"×6.0") and is sized so as not to obscure a first license plate display region **162** (e.g., a license plate number) for a United States vehicle license plate. In some examples, the example central opening **142** is less than about 520 mm×110 mm (20.5"×4.3") and is sized so as not to obscure a first license plate display region **162** (e.g., a license plate number) for a vehicle license plate in some European countries. Likewise, in other examples, the example central opening **142** is less than about 305 mm×152 mm (12.0"×6.0"), 305 mm×160 mm (12.0"×6.3"), 372 mm×135 mm (14.6"×5.3"), 440 mm×120 mm (17.3"×4.7"), 300 mm×80 mm (11.8"×3.1"), or 440 mm×140 mm (17.3"×5.5").

The example frame **140** further defines one or more example frame display regions **146** adjacent the example central opening **142**. In the example of FIG. 1, the example frame **140** is shown to define a plurality of frame display regions **146**, more particularly, four example frame display regions **146**. The example frame **140** of FIG. 1 includes an example bottom frame display region **146** below an example bottom edge of the central opening **142**, an example first lateral frame display region **146** adjacent an example first lateral edge of central opening **142**, an example second lateral frame display region **146** adjacent an example second lateral edge of central opening **142**, or an example upper frame display region **146** above an example top edge of the central opening **142**. The one or more example frame display regions **146** include text and/or graphics, which are generically represented in FIG. 1 by fields or arrays of "X" indicia. In some examples, one or more of the example frame display region(s) **146** include(s) user-selectable text or graphics (e.g., selected from a list of options provided on a website, custom printed, etc.), such as may be associated with sports team logos, organizations, colleges or universities, companies, phrases, causes, etc.

The example frame **140** of FIG. 1 defines a plurality of contiguous walls including an example first wall **141**, an example second wall **145**, an example third wall **148** and an example fourth wall **149**. The example walls **141**, **145**, **148**, **149** extend forwardly to define a first volume (not numbered) at the front of the example frame **140** to receive the example transmissive substrate **120**. The example walls **141**, **145**, **148**, **149** also extend rearwardly to define a second volume (not numbered) at the rear of the example frame **140** to receive the example license plate **160** and at least a portion of the example base **200**. In some examples, the example walls **141**, **145**, **148**, **149** extend both forwardly and

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rearwardly. In some examples, a first set of walls extends forwardly and a second set of walls extends rearwardly.

While a plurality of example walls **141**, **145**, **148**, **149** are shown in the example frame **140** of FIG. 1, a lesser number of walls could be provided and/or the configuration of the walls altered. For instance, while not a presently preferred example, the example walls **141**, **145**, **148**, **149** could extend rearwardly, but not forwardly. Instead of the example transmissive substrate **120** being received within the first volume defined by the example walls **141**, **145**, **148**, **149**, the example transmissive substrate **120** could instead positively connect to the example frame **140** via a plurality of male (or female) connectors (e.g., snap fit connectors, etc.) formed on the example transmissive substrate **120** to matingly engage with correspondingly configured and positioned female (or male) connectors formed on the example frame **140**. In another example, the example transmissive substrate **120** is connected to the example frame **140** via a plurality of mechanical fasteners (e.g., screws, etc.).

In yet another example, the frame **140** could include a subset of the walls of FIG. 1, such as only wall **149** or only wall **148**. In such example, the example transmissive substrate **120** could positively connect to the example frame **140** via a plurality of male (or female) connectors (e.g., snap fit connectors, etc.) formed on the example transmissive substrate **120** to matingly engage with correspondingly configured and positioned female (or male) connectors formed on the example frame **140** or, could positively connect to the example frame **140** via a plurality of mechanical fasteners (e.g., screws, etc.).

The example frame **140** includes one or more front connector(s) to removably attach the example transmissive substrate **120** to the front of the example frame **140**. It the example of FIG. 1, the example frame **140** includes four example front connectors **147** disposed adjacent a front of the example frame **140**. The four example front connectors **147** are cantilevered snap-fit connectors, having a head that is chamfered or tapered at a top surface, to facilitate biasing of the heads away from the first volume upon movement of the example transmissive substrate **120** into the first volume, wherein lateral edges of the example transmissive substrate **120** contact the chamfered or tapered top surfaces of the heads of the example front connectors **147** and slightly bias the heads outwardly. Following passage of the example transmissive substrate **120** into the first volume and past a bottom surface of the head, where it no longer outwardly biases the heads of the example front connectors **147**, the heads returns to their initial positions and lower surfaces of the heads facilitate retention of the example transmissive substrate **120** within the first volume. For instance, in some examples, the lower surfaces of the heads of the example front connectors **147** are at flat and are at least substantially parallel to the front surface of the example transmissive substrate **120** such that the heads would have to be manually or digitally biased outwardly to facilitate removal of the example transmissive substrate **120**. While four example front connectors **147** are shown, a greater number (e.g., five, six, etc.) or a lesser number (e.g., one, two, three) could alternatively be implemented in the example frame **140**. Likewise, different forms of example front connectors **147** (e.g., male/female connectors, mechanical connectors, fasteners, etc.) may be implemented to facilitate removable attachment of the example transmissive substrate **120** to the example frame **140**.

In some examples, the example frame **140** includes example openings **144**. In some examples of the present concepts, the example openings **144** are omitted.

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FIG. 1 also shows an example license plate **160**, which may be formed from a variety of materials, such as plastic or metal (e.g., aluminum, steel, an alloy, etc.) or, in some instances, from card stock (e.g., a temporary tag). The example license plate **160** includes an example first license plate display region **162** for display of a license plate number and additional relevant information (e.g., a state of issuance, a state motto, etc.) and an example second license plate display region **164** for display of an emblem or a sticker (e.g., a governmental vehicle registration sticker, etc.) that may be required in a given jurisdiction. In some examples, a third license plate display region (not shown) may be required, similar to that of the example second license plate display region **164**, in jurisdictions requiring a second emblem or a sticker. The example license plate **160** includes a plurality of example through holes **165** to receive example mechanical fasteners **150** (e.g., screws, etc.) to secure the example license plate **160** to the example base **200** and/or to the vehicle itself which bears the example license plate holder **100**.

As discussed further in relation to FIGS. 2-7B below, the example frame **140** includes one or more rear connectors (not shown in FIG. 1) disposed adjacent a rear of the example frame **140** to removably attach the rear of the example frame **140** to the example base **200**.

As noted above, the example license plate **160** may include any dimension specified by any jurisdiction including, but not limited to, for example, 305 mm×152 mm (12.0"×6.0"), 520 mm×110 mm (20.5"×4.3"), or 440 mm×140 mm (17.3"×5.5") to name but a few examples.

The example base **200** of FIG. 1 includes an example central section **202**, an example first lateral section **203**, and an example second lateral section **204**. Formed within the example central section **202** are one or more example first connectors **205** (e.g., through holes, bosses, male connectors, female connectors, etc.) for attaching the example base **200** to a vehicle. The example central section **202** is recessed relative to the example first lateral section **203** and the example second lateral section **204** to define a recessed volume within which the example license plate **160** may be received. The recessed volume of the example central section **202** has an area sufficient to receive an example license plate **160** of a jurisdiction of relevance (e.g., greater than about 305 mm×152 mm (12.0"×6.0") in the United States, greater than about 440 mm×140 mm (17.3"×5.5") in China, etc.). In some examples, the example central section **202**, the example first lateral section **203**, and the example second lateral section **204** are substantially planar and the example first lateral section **203** and the example second lateral section **204** do not define a recessed volume therebetween.

The exploded view of FIG. 1 shows, beneath the example base **200**, example rear connectors **210** (e.g., mechanical fasteners, etc.) and example adapters **215** (e.g., anchors, anti-rotation anchors, etc.). The example rear connectors **210**, discussed further below in relation to FIGS. 3 and 5-6, are inserted through corresponding openings in the example frame **140** and the example base **200** to secure a lower portion of the example frame **140** to a lower portion of the example base **200**. In some examples, the example adapters **215** are inserted only into the example base **200**. In some examples, the example adapters **215** are inserted through the example frame **140** and into the example base **200**. In some examples, the example adapters **215** are omitted.

FIG. 2A is a perspective view of the example license plate holder **100** of FIG. 1 in an assembled condition and FIG. 2B is a front view of the example license plate holder **100** of FIG. 1 in an assembled condition. FIGS. 2A-2B show the

alignment of the example transmissive substrate 120 within the first volume of the example frame 140 and the alignment of the example frame 140 relative to the example license plate 160 and the example base 200 (not shown in FIGS. 2A-2B). FIGS. 2A-2B show that the example frame display regions 146 are viewable through, and protected by, the example transmissive substrate 120. FIGS. 2A-2B also show that the example first license plate display region 162 is viewable through the example first region 126 of the example transmissive substrate 120 and the example central opening 142 of the example frame 140. Likewise, FIGS. 2A-2B also show that the example second license plate display region 164 is viewable through the example third region 128 of the example transmissive substrate 120 and the example secondary opening 143 of the example frame 140.

FIG. 3 is an example bottom view of the assembled example license plate holder 100 of FIGS. 2A-2B, with the example front 300 of the example frame 140 and the example rear 302 of the example frame 140 being shown. FIG. 3 shows an example first boss 310 formed on a first side of the example wall 149 or bottom portion of the example frame 140 and an example second boss 312 formed on a second side of the example wall 149 of the example frame 140. FIG. 3 also shows an example first boss 310 formed on a first side of the example wall 149 or bottom portion of the example frame 140 and an example second boss 312 formed on a second side of the example wall 149 of the example frame 140. The example first boss 310 aligns with (e.g., is at least substantially coaxial with) an example third boss 615 (not shown in FIG. 3) formed on a first side of an example bottom portion 249 of the example base 200. The example second boss 312 aligns with (e.g., is at least substantially coaxial with) an example fourth boss 616 (not shown in FIG. 3) formed on a second side of an example bottom portion 249 of the example base 200. Each of the example first boss 310 and the example second boss 312 include an example through hole 311 to facilitate receipt of the example mechanical fastener 250, which connects the example first boss 310 and the example second boss 312 of the example frame 140 with, respectively, the example third boss 615 and the example fourth boss 616 of the example base 200.

FIG. 4 is an example top view of the assembled example license plate holder 100 of FIGS. 2A-3. FIG. 4 shows a top view of an example first upper rear connector 410 and an example second upper rear connector 412, which are shown more fully in FIGS. 5-6.

FIG. 5 is an example rear perspective view of the example license plate holder 100 of FIGS. 1-4 in the assembled condition and FIG. 6 is an example rear view of the example license plate holder of FIGS. 1-5. At the upper end of the assembled example license plate holder 100, FIGS. 5-6 show the example first upper rear connector 410 engaging the example ledge 710 of the example first lateral section 203 of the example base 200. FIGS. 5-6 also show the example second upper rear connector 412 engaging the example ledge 712 of the example second lateral portion 204 of the example base 200. In some examples, the example ledges 710, 712 correspond to an upper portion of a recess formed in a rear portion of the example first lateral section 203 and example second lateral portion 204, respectively. In other examples, the rear surface of the example base 200 may be substantially planar and the example ledges 710, 712 may be formed to extend rearwardly therefrom.

Even in isolation (i.e., without the example rear connectors 210), the example first upper rear connector 410 and the

example second upper rear connector 412 provide positive engagement with, and lockup of, the example frame 140 and the example base 200.

In FIGS. 5-6, the example mechanical fasteners 150 (e.g., screws, etc.) are shown to protrude through the example first connectors 205 (e.g., through holes, bosses, male connectors, female connectors, etc.) of the example base 200 and, in an installed condition, would connect the example license plate holder 100 to a vehicle.

The right example rear connector 210, in the view of FIGS. 5-6, comprises a mechanical fastener (e.g., a screw, etc.) inserted through the example through hole 311 of the example first boss 310 of the example frame 140 and through a corresponding one of the example adapters 215, which is in turn through or received by the example third boss 615 of the example base 200. The left example rear connector 210, in the view of FIGS. 5-6, comprises a mechanical fastener (e.g., a screw, etc.) inserted through the example through hole 313 of the example second boss 312 of the example frame 140 and through a corresponding one of the example adapters 215, which is itself inserted into or received by the example fourth boss 616 of the example base 200.

While the view of FIGS. 5-6 shows the example first lateral section 203 and the example second lateral section 204 of the example base 200 to define recesses on the rear of the example base 200, in other examples the example first lateral section 203 and the example second lateral section 204 are solid and the example third boss 615, the example fourth boss 616, and the example adapters 215 are omitted in favor of holes (e.g., threaded holes, etc.) to receive the example rear connectors 210. In some examples, rather than two example rear connectors 210 disposed at or near a first and a second lateral end of the example frame 140 and the example base 200, a single centrally disposed rear connector 210 is used to secure a lower portion of the example frame 140 to a lower portion of the example base 200.

FIGS. 7A-7B are, respectively, example right side and left side views of the example license plate holder 100 of FIGS. 1-6, showing alternate views of components of the example frame 140 including the example first upper rear connector 410, the example second upper rear connector 412, the example first boss 310 and the example second boss 312.

FIGS. 8-15B are views of a second embodiment of an example license plate holder in accord with at least some of the present concepts, such second embodiment including some similarities with the example license plate holder of FIGS. 1-7B. The example second embodiment of the example license plate holder shown in FIGS. 8-15B includes, for example, a different mechanical connection between the example frame 1140 and the example base 1200 than that of the example of FIGS. 1-7B and a different mode of attachment of the example base 1200 to the example frame 1140. In the example license plate holder of FIGS. 8-15B, the example frame 1140 is positioned laterally off-center relative to the example base 1200 during assembly so that example first rear connectors 1500 on the example base 1200 (see, e.g., FIG. 8) are positioned to move laterally relative to example second rear connectors 1600 on the example frame 1140 (see, e.g., FIG. 9) from an initial mounting position to a final assembled position during assembly and the example first rear connectors 1500 and the example second rear connectors 1600 are positioned for securement relative to one another in a final, assembled position (e.g., centered) of the example frame 1140 relative to the example base 1200. In some examples, the example first rear connectors 1500 abut against and frictionally

engage the example second rear connectors **1600** in the final, assembled position to restrict relative motion therebetween. In some examples, the example first rear connectors **1500** interlock with the example second rear connectors **1600** in the final, assembled position, and optionally include one or more latching elements. In some examples, one or more of the example first rear connectors **1500** include one of a male or female connection member and one or more of the example second rear connectors **1600** include another one of a male or female connection member so that the example first rear connectors **1500** lock into the example second rear connectors **1600** in the final, assembled position.

In some examples, one or more of the example first rear connectors **1500** are not as illustrated, but are rather a conventional male or female connection member and one or more of the example second rear connectors **1600** are not as illustrated, but are another one of a male or female connection member so that such first rear connector locks into the second rear connector **1600** in the final, assembled position. The example first rear connectors **1500** and the example second rear connectors **1600** may comprise any complementary shape adapted for engagement such as, but not limited to, complementary right prisms (e.g., wedge) leading to a plateau (e.g., a rectangular shape or square shape having a height equal to that of a base of the right prism), or a rack and pinion arrangement, to name but two examples.

While a certain number of and arrangement of example first rear connectors **1500** and example second rear connectors **1600** are shown in the example license plate holder **1100** of FIGS. 8-15B, the present concepts include other numbers of and/or arrangement of the example first rear connectors **1500** and example second rear connectors **1600**. For instance, the present concepts include one, two, three or more sets of a first rear connector **1500** and a second rear connector **1600**. The one, two, three or more sets of the first rear connector **1500** and the second rear connector **1600** may be arranged asymmetrically (e.g., a different number of and/or position sets of one or more first rear connectors and one or more second rear connectors as between a top and bottom of the example license plate holder, etc.) or on different surfaces of the example base **1200** and example frame **1140**.

In some examples, the example first rear connectors **1500** and the example second rear connectors **1600** are positioned on vertical or lateral surfaces of the example frame **1140** and the example base **1200** (e.g., left surfaces and right surfaces in the orientation shown) to facilitate mounting of the frame **1140** to the base **1200** by positioning the frame **1140** vertically off-center relative to the base **1200** during assembly and moving the frame **1140** relative to the base **1200** to center the frame **1140** relative to the base **1200** in the final assembled position. In this configuration, during assembly, the second rear connectors **1600** on the frame **1140** move vertically relative to the first rear connectors **1500** on the base **1200** as the frame **1140** is moved from the initial mounting position to a final assembled position. In the final assembled position, the first rear connectors **1500** and the second rear connectors **1600** engage one another.

Further, while the example second embodiment of the example license plate holder shown in FIGS. 8-15B shows the example frame **1140** and the example base **1200** being connected at the bottom via two mechanical fasteners (e.g., screws), other variants of the designs of FIGS. 1-7B and of 8-15B optionally reduce the number of mechanical fasteners (e.g., only one screw, only one pin, etc.) or omit the use of mechanical fasteners in favor of complementary mechanical

connectors (e.g., clasps, snap-fit connectors, male/female connectors, etc.) formed in the example frame and the example base.

FIG. 8 is an example exploded front perspective view of the example second embodiment of an example license plate holder **1100** in accord with at least some of the present concepts. The example license plate holder **1100** is shown to include, from left-to-right, an example transmissive substrate **1120**, an example frame **1140** and an example base **1200**. An optional example license plate **1160** is also shown in relation to the example license plate holder **1100** for contextual illustration to show an example integration of a license plate **1160** with a license plate holder **1100** in accord with at least some aspects of the present concepts.

The example transmissive substrate **1120** may include, for instance, but is not limited to, a substrate of glass, treated or strengthened glass (e.g., Gorilla Glass (Corning), Dragontrail (AGC Inc.), Xensation (Schott AG)), Plexiglass (e.g., Lucite, Acrylite, Perspex, Oroglass Optix, Altuglass, etc.), polycarbonate, acrylic (e.g., injection molded acrylic or polymethyl methacrylate, polystyrene or PMMA. In some examples, the example transmissive substrate **1120** is a uniform substrate of a selected material. In some examples, the example transmissive substrate **1120** defines a plurality of different regions such as an example first region **1126** (e.g., a central region) having a first set of characteristics (e.g., optical properties such as refractive index, transmission/transmittance, absorption, surface finish, graphics, mechanical properties such as Young's modulus, strength, hardness, impact resistance, chemical resistance, etc.), an example second region **1125** (e.g., a peripheral region) having a second set of characteristics, an example third region **1128** having a third set of characteristics, etc. Thus, in various examples, the example transmissive substrate **1120** can have a single uniform region (i.e., uniform characteristics at least substantially throughout) or a plurality of different regions (e.g., two regions, three regions, four regions, or more than four regions) having different characteristics (e.g., different graphics, etc.). In some examples, two regions may have the same characteristics (e.g., the example first region **1126** and the example second region **1125** both having the same optical properties). In some examples, one or more of the regions could comprise an opening.

In the example of FIGS. 8-15B, the example transmissive substrate **1120** comprises a substrate of glass that is about 0.12 inches thick. In other examples, the example transmissive substrate **1120** comprises a substrate of glass that is less than 0.12 inches thick or greater than 0.12 inches thick, such as between 0.12-0.25 inches thick. In some examples, the example transmissive substrate **1120** is uniform, with substantially uniform characteristics throughout. In some examples, the example transmissive substrate **1120** defines a plurality of different regions, such as an example first region **1126** (e.g., a central region) having a first set of characteristics (e.g., a first set of optical properties) and an example second region **1125** (e.g., a peripheral region) having a second set of characteristics (e.g., a second set of optical properties). In one example, for instance, the example first region **1126** is clear and the example second region **1125** is slightly opaque. In another example, the example first region **1126** is clear and the example second region **1125** includes paint, metallization, a silkscreen, or an appliqué (e.g., a sticker). In some examples, the example first region **1126** and the example second region **1125** have the same characteristics. It is to be noted that, while the example second region **1125** (e.g., a peripheral region) is shown to be

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symmetric, the example second region **1125** may be asymmetric, non-uniform or stylized (e.g., a silk-screened second region **1125** defining a first region **1126** that is a transparent rectangle offset at an angle relative to the perimeter of the example transmissive substrate **1120**, a silk-screened second region **1125** that depicts flames or waves, etc.).

In the example transmissive substrate **1120** shown in FIG. **8**, the example first region **1126** generally corresponds in size to a display area (e.g., a DIN 1451 license plate number) of an example license plate (e.g., example license plate **1160**) to be incorporated into the example license plate holder **1100** (e.g., a United States license plate, a European license plate, a Chinese license plate, etc.). For instance, in some examples, the example first region **1126** is less than 12.0"×6.0" (305 mm×152 mm) for a United States license plate, less than 20.5"×4.3" (520 mm×110 mm) for a license plate of some European countries, or 17.3"×5.5" (440 mm×140 mm) for a Chinese license plate. In some examples, the example first region **1126** is a cutout in the example transmissive substrate **1120**. In some examples, the example first region **1126** includes an example lateral region **1128** to facilitate display of an emblem or sticker (e.g., a governmental vehicle registration sticker, tax stamp etc.), such lateral region **1128** being sized and located in a position required by the local jurisdiction. In some examples, the example lateral region **1128** is a different size (e.g., larger, smaller, etc.) and/or location than that shown in the example transmissive substrate **1120** of FIG. **8** to comport with sizing and/or location requirements of any particular jurisdiction for the display of an emblem or sticker (e.g., a governmental vehicle registration sticker, tax stamp etc.). For instance, in the example of FIG. **8**, the symmetry of the example transmissive substrate **1120** provides regions corresponding to the example lateral region **1128** on the upper left side, as well as on the bottom left and bottom right, of the example transmissive substrate **1120**.

The example frame **1140** defines an example central opening **1142** to permit display of indicia, such as the license plate number, on an example first license plate display region **1162**. In some examples, such as is shown in FIG. **8**, the example central opening **1142** is a singular opening that reveals, through the example central opening **1142**, all portions of the underlying example license plate **1160** including the example first license plate display region **1162**, the example second license plate display region **1164**, the example third license plate display region **1161A**, and the example fourth license plate display region **1161B**.

In other examples, the example central opening **1142** may alternatively be sized differently (e.g., larger, smaller, etc.) to comport with sizing and/or location requirements of any particular jurisdiction and/or to selectively reveal or obscure, for instance, the example third license plate display region **1161A** and/or the example fourth license plate display region **1161B**.

In the example of FIGS. **8-15B**, the example second region **1125** of the example transmissive substrate **1120** defines a plurality of example salients **1122** extending inwardly toward the example first region **1126**. In some examples, these example salients **1122** are sized and spaced so as to partially or fully visually obscure one or more of (e.g., some of or all of) the example mechanical fasteners **1150** used to secure the example license plate **1160** to the example base **1200** and/or vehicle. The example salients **1122** may have the same characteristics as the example second region **1125** (e.g., a second set of optical properties) or different characteristics from the example second region **1125** (e.g., a third set of optical properties). In some

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examples, a shape and/or characteristic of the example salients **1122** may reflect a selected theme of the example license plate holder **1100**. In one example, for instance, the example first region **1126** is clear and the example second region **1125** includes graphics evocative of a beach theme, with the example salients **1122** taking the form of an ellipse (e.g., graphics of a surf board in sand) or a semi-circle (e.g., graphics of a beach ball). As another example, the example first region **1126** is clear and the example second region **1125** includes graphics evocative of a sports theme, the example salients **1122** taking the form of an ellipse (e.g., graphics of a football) or a semi-circle (e.g., graphics of a soccer ball). It is to be noted that the example salients **1122** may comprise a uniform shape or form, such as is shown in the example of FIG. **8**, or may comprise different shapes or forms (e.g., two, three, four or more different shapes, etc.) adapted to partially or fully visually obscure one or more of (e.g., some of or all of) the example mechanical fasteners **1150** used to secure the example license plate **1160** to the example base **1200** and/or vehicle.

In some examples, the example central opening **1142** of the example frame **1140** is less than about 305 mm×152 mm (12.0"×6.0") and is sized so as not to obscure a first license plate display region **1162** (e.g., a license plate number) for a United States vehicle license plate. In some examples, the example central opening **1142** is less than about 520 mm×110 mm (20.5"×4.3") and is sized so as not to obscure a first license plate display region **1162** (e.g., a license plate number) for a vehicle license plate in some European countries. Likewise, in other examples, the example central opening **1142** is less than about 305 mm×152 mm (12.0"×6.0"), 305 mm×160 mm (12.0"×6.3"), 372 mm×135 mm (14.6"×5.3"), 440 mm×120 mm (17.3"×4.7"), 300 mm×80 mm (11.8"×3.1"), or 440 mm×140 mm (17.3"×5.5").

The example frame **1140** further defines one or more example frame display regions **1146** adjacent the example central opening **1142**. In the example of FIG. **8**, the example frame **1140** is shown to define a plurality of frame display regions **1146**, more particularly, four example frame display regions **1146**. The example frame **1140** of FIG. **8** includes an example bottom frame display region **1146** below an example bottom edge of the example central opening **1142**, an example first lateral frame display region **1146** adjacent an example first lateral edge of example central opening **1142**, an example second lateral frame display region **1146** adjacent an example second lateral edge of example central opening **1142**, or an example upper frame display region **1146** above an example top edge of the example central opening **1142**. The one or more example frame display regions **1146** include text and/or graphics, which are generically represented in FIG. **8** by fields or arrays of "X" indicia. In some examples, one or more of the example frame display region(s) **1146** include(s) user-selectable text or graphics (e.g., selected from a list of options provided on a website, custom printed, etc.), such as may be associated with sports team logos, organizations, colleges or universities, companies, phrases, causes, etc.

In another example, the one or more example frame display regions **1146** are omitted and, instead, one or more example transmissive substrate **120**, **1120** display regions (not shown) are provided with text and/or graphics (e.g., printed on a front side and/or a back side of the transmissive substrate **120**, **1120**, an appliqué on a front side and/or a back side of the transmissive substrate **120**, **1120**, etc.). In some examples, such text and/or graphics provided on one or more regions of the example transmissive substrate **120**, **1120** includes user-selectable text or graphics

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(e.g., selected from a list of options provided on a website, custom printed, etc.), such as may be associated with sports team logos, organizations, colleges or universities, companies, phrases, causes, etc.

The example frame **1140** of FIG. **8** defines a plurality of contiguous walls including an example first wall **1141**, an example second wall **1145**, an example third wall **1148** and an example fourth wall **1149**. The example walls **1141**, **1145**, **1148**, **1149** extend forwardly to define a first volume (not numbered) at the front of the example frame **1140** to receive the example transmissive substrate **1120**. The example walls **1141**, **1145**, **1148**, **1149** also extend rearwardly to define a second volume (not numbered) at the rear of the example frame **1140** to receive the example license plate **1160** and at least a portion of the example base **1200**. In some examples, the example walls **1141**, **1145**, **1148**, **1149** extend both forwardly and rearwardly. In some examples, a first set of walls extends forwardly and a second set of walls extends rearwardly.

While a plurality of example walls **1141**, **1145**, **1148**, **1149** are shown in the example frame **1140** of FIG. **8**, a lesser number of walls could be provided and/or the configuration of the walls altered. For instance, while not a presently preferred example, the example walls **1141**, **1145**, **1148**, **1149** could extend rearwardly, but not forwardly, and instead of the example transmissive substrate **1120** being received within the first volume defined by the example walls **1141**, **1145**, **1148**, **1149**, the example transmissive substrate **1120** could instead positively connect to the example frame **1140** via a plurality of mechanical connectors (e.g., male/female connectors) formed on the example transmissive substrate **1120** to matingly engage with correspondingly configured and positioned female (or male) connectors formed on the example frame **1140** and/or via a plurality of mechanical fasteners (e.g., screws, etc.).

In yet another example, the frame **1140** could include a subset of the walls of FIG. **8**, such as only wall **1149** or only wall **1148**. In such example, the example transmissive substrate **1120** could positively connect to the example frame **1140** (e.g., to the wall **1149** or the wall **1148**, etc.), via a plurality of male (or female) connectors (e.g., snap fit connectors, etc.) formed on the example transmissive substrate **1120** to matingly engage with correspondingly configured and positioned female (or male) connectors formed on the example frame **1140** and/or via a plurality of mechanical fasteners (e.g., screws, etc.).

The example frame **1140** includes one or more example front connector(s) **1147** to removably attach the example transmissive substrate **1120** to the front of the example frame **1140**. In the example of FIG. **8**, the example frame **1140** includes four example front connectors **1147** disposed adjacent a front of the example frame **1140**. The four example front connectors **1147** are cantilevered snap-fit connectors, having a head that is chamfered or tapered at a top surface, to facilitate biasing of the heads away from the first volume upon movement of the example transmissive substrate **1120** into the first volume, wherein lateral edges of the example transmissive substrate **1120** contact the chamfered or tapered top surfaces of the heads of the example front connectors **1147** and slightly bias the heads outwardly. Following passage of the example transmissive substrate **1120** into the first volume and past a bottom surface of the head, where it no longer outwardly biases the heads of the example front connectors **1147**, the heads return to their initial positions and lower surfaces of the heads facilitate retention of the example transmissive substrate **1120** within the first volume. For instance, in some examples, the lower

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surfaces of the heads of the example front connectors **1147** are at flat and are at least substantially parallel to the front surface of the example transmissive substrate **1120** such that the heads would have to be manually or digitally biased outwardly to facilitate removal of the example transmissive substrate **1120**. While four example front connectors **1147** are shown, a greater number (e.g., five, six, etc.) or a lesser number (e.g., one, two, three) could alternatively be implemented in the example frame **1140**. Likewise, different forms of example front connectors **1147** (e.g., male/female connectors, mechanical connectors, fasteners, etc.) may be implemented to facilitate removable attachment of the example transmissive substrate **1120** to the example frame **1140**.

FIG. **8** also shows an example license plate **1160**, which may be formed from a variety of materials, such as plastic or metal (e.g., aluminum, steel, an alloy, etc.) or, in some instances, from card stock (e.g., a temporary tag). The example license plate **1160** includes an example first license plate display region **1162** for display of a license plate number and additional relevant information (e.g., a state of issuance, a state motto, etc.) and an example second license plate display region **1164** for display of an emblem or a sticker (e.g., a governmental vehicle registration sticker, etc.) that may be required in a given jurisdiction. In some examples, the example license plate **1160** includes a third license plate display region (e.g., **1161A**) and/or a fourth license plate display region (e.g., **1161B**).

The example license plate **160** includes a plurality of example through holes **1165** to receive example mechanical fasteners **1150** (e.g., screws, etc.) to secure the example license plate **1160** to the example base **1200** and/or to the vehicle itself which bears the example license plate holder **1100**.

As discussed further in relation to FIGS. **9A-15B** below, the example frame **1140** includes one or more rear connectors (not shown in FIG. **8**) disposed adjacent a rear of the example frame **1140** to removably attach the rear of the example frame **1140** to the example base **1200**.

As noted above, the example license plate **1160** may include any dimension specified by any jurisdiction including, but not limited to, for example, 305 mm×152 mm (12.0"×6.0"), 520 mm×110 mm (20.5"×4.3"), or 440 mm×140 mm (17.3"×5.5") to name but a few examples.

The example base **1200** of FIG. **8** includes an example central section **1202** having one or more example first connectors **1205** (e.g., through holes, bosses, male connectors, female connectors, etc.) for attaching the example base **1200** to a vehicle. The example central section **1202** optionally defines a recessed volume within which the example license plate **1160** may be received or a raised volume against which the example license plate **1160** may be secured. For instance, the recessed volume of the example central section **1202** has an area sufficient to receive an example license plate **1160** of a jurisdiction of relevance (e.g., greater than about 305 mm×152 mm (12.0"×6.0") in the United States, greater than about 440 mm×140 mm (17.3"×5.5") in China, etc.). In some examples, the example central section **1202** is substantially planar.

The exploded view of FIG. **8** shows, at a bottom portion of the example frame **1140**, example mechanical fasteners **1210**, discussed further below in relation to FIGS. **9-15 B**, which are inserted through corresponding openings (not shown in FIG. **8**) in the example frame **1140** and openings (not shown in FIG. **8**) in the example base **200** to secure a lower portion of the example frame **1140** to a lower portion of the example base **1200**.

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FIG. 9A is an example rear exploded perspective view of the example license plate holder of FIG. 8. Further to FIG. 8, FIG. 9A shows a plurality of example first rear connectors 1500 on upper and lower positions of the example base 1200 and a corresponding plurality of example second rear connectors 1600 on upper and lower positions of the example frame 1140. FIG. 9A also shows a plurality of example stops 1151 at a left-hand side of the frame, when viewed from the rear. As noted above, when the example frame 1140 of FIGS. 8-15B is mounted to the example base 1200 after the example base 1200 is fixed in place (e.g., to a vehicle), the example frame 1140 is positioned laterally off-center from the example base so that the example first rear connectors 1500 and the example second rear connectors 1600 are similarly laterally off-center from one another and the example first rear connectors 1500 and the example second rear connectors 1600 engage one another as the example frame 1140 is translated (e.g., toward the right) until a rightmost edge of the example license plate 1160 or the example base 1200 contacts the example stops 1151 at a centered position of the example frame 1140.

It is to be noted that both the placement of, and the number of, the example first rear connectors 1500 and the example second rear connectors 1600 may be varied and may include a lesser number (e.g., one or more sets of the example first rear connector 1500 and the example second rear connector 1600) or a greater number than that illustrated. Additionally, while one example manner of example first rear connector 1500 and example second rear connector 1600 are depicted, the physical makeup of the example first rear connector 1500 and example second rear connector 1600 may manifest in other form factors, such as a male connector and a female connector or latches. In some examples, the example first rear connector 1500 and example second rear connector 1600 may comprise magnetic connectors (e.g., neodymium magnets). In some examples, the example license plate holder 1100 may include a first set of an example first rear connector 1500 and example second rear connector 1600 that is different in structure and/or operation than a second set of an example first rear connector 1500 and example second rear connector 1600. In other words, the example first rear connectors 1500 and the example second rear connectors 1600 need not be identical.

FIG. 9B shows an example rear close-up view of a portion of the example base 1200 of FIG. 9A showing an example first rear connector 1500. The example first rear connector 1500 of FIG. 9B includes an example first portion 1502 that includes a substantially flat sliding or bearing surface (e.g., positioned in a middle portion of the example first portion 1502) and an example second portion 1504 that includes one or more example protuberances (e.g., ribs, projecting members, etc.). In some examples, the example second portion 1504 is omitted and the example first rear connector 1500 includes a substantially flat sliding or bearing surface along substantially an entire length thereof.

FIG. 10A is a front perspective view of the example license plate holder of FIGS. 8-9B in an assembled state. FIG. 10B is a front view of the example license plate holder of FIGS. 8-10A in an assembled state. FIGS. 10A-10B show the alignment of the example transmissive substrate 1120 within the first volume of the example frame 1140 and relative to the example front connectors 1147. FIGS. 10A-10B also show the alignment of the example transmissive substrate 1120 relative to the example first license plate display region 1162 and the example second license plate display region 1164 of the example license plate 1160, such

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regions being viewable through the example lateral region 1128 of the example transmissive substrate 1120 and the example central opening 1142 of the example frame 1140, respectively.

FIGS. 11-12 are an example bottom view and an example top view, respectively, of the example license plate holder 1100 of FIGS. 8-10B in an assembled state, with the example front 1300 of the example frame 1140 and the example rear 1302 of the example frame 1140 being shown. The example frame 1140 is shown to include two example regions openings (not numbered) in which example mechanical fasteners 1210 (e.g., screws, etc.) are disposed to pass through corresponding positioned and dimensioned openings in the example base 1200 to enhance securement of the example frame 1140 to the example base 1200. Example mechanical fasteners 1150, which connect the example license plate holder 1100 to a vehicle, are shown to extend from the rear of the example base 1200.

FIG. 13 is an example rear perspective view of the example license plate holder 1100 of FIGS. 8-12 in an assembled state. FIGS. 14A-14B are, respectively, an example rear view of the example license plate holder 1100 of FIGS. 8-13 in an assembled state and a close-up rear view of a portion of the example license plate holder 1100 of FIG. 14A showing an example connection between components of the example license plate holder 1100. At an example upper end and at an example lower end of the assembled example license plate holder 1100, FIGS. 13-14B show the example first rear connectors 1500 on the example base 1200 engaged (e.g., interlocking, frictional engagement, abutting, etc.) with the example second rear connectors 1600 on the example frame 1140 to resist lateral translation and rotation of the example frame 1140 relative to the example base 1200. In the assembled views of FIGS. 13-14B, the lateral edge of the example base 1200 can be seen to abut against the example stop(s) 1151 of the example frame 1140. In some examples, the example license plate holder 1100 omits the example stop(s) 1151 entirely. In such example, the example license plate holder 1100 may optionally utilize of the example second portion 1504 of one or more of the example first rear connectors 1500 as a mechanical stop.

In the views of FIGS. 13-14B, the example mechanical fasteners 1210 (e.g., screws, etc.) shown in FIGS. 9A and 11 are disposed within the example first boss 1616 and the example second boss 1615 formed in the example base 1200 to secure the example frame 1140 to the example base 1200. In some examples, only one example mechanical fastener 1210 is used to secure the example frame 1140 to the example base 1200. In some examples the example mechanical fastener(s) 1210 are optionally omitted in favor of a more positive lockup between one or more of the example first rear connectors 1500 and a corresponding one or more of the example second rear connectors 1600 (e.g., a locking male/female connector, etc.). In other examples, the example mechanical fastener 1210 could comprise a pin or dowel disposed through the example frame 1140 and into the example base 1200 to prevent relative motion of the example frame 1140 relative to the example base and/or the example license plate 1160, such as a pin inserted through an upper portion of the example frame 1140 and into the example base, such pin being retained in place by gravity and/or an adhesive (e.g., Loctite, etc.). In other examples, the mechanical fastener 1210 could comprise one or more springs, or a mechanical insert, to bias a lateral edge of the example base 1200 or a lateral edge of the example license plate 1160 against the example stops 1151 of the example frame 1140.

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FIGS. 15A-15B are, respectively, example right side and left side views of the example license plate holder 1100 of FIGS. 8-14B in an assembled state, showing alternate views of components of the example frame 1140 and the example base 1200.

FIG. 16 is an example front perspective view of an example third embodiment of an example license plate holder 2000 in an assembled state in accord with at least some of the present concepts. FIG. 16 shows an example alignment of the example license plate transmissive substrate 1120 relative the example frame 2140. In the example of FIG. 16, the example license plate transmissive substrate 1120 is adhered to the example frame 2140 via an optically clear adhesive. In accord with other examples, the example license plate transmissive substrate 1120 is connected to the example frame 2140 via one or more mechanical connection members, such as via example front connectors 1147. FIG. 16 also shows an example alignment of the example transmissive substrate 1120 relative to the example first license plate display region 1162, the example second license plate display region 1164, the example third license plate display region 1161A, and the example fourth license plate display region 1161B, such regions being viewable through the example first region 1126 (e.g., central region) and the example third region 1128 of the example transmissive substrate 1120 and through a corresponding opening of the example frame 2140.

Similar to the preceding example of FIGS. 8-15B, the example third embodiment of an example license plate holder 2000 of FIGS. 16-19 also positions the example frame 2140 laterally off-center relative to the example base 2200 during assembly so that example first rear connectors 2500 on the example base 2200 (see, e.g., FIG. 17) are positioned to move laterally relative to example second rear connectors 2600 on the example frame 2140 (see, e.g., FIG. 18) from an initial mounting position to a final assembled position during assembly and the example first rear connectors 2500 and the example second rear connectors 2600 are positioned for securement relative to one another in a final, assembled position (e.g., centered) of the example frame 2140 relative to the example base 2200. In some examples, the example first rear connectors 2500 abut against and frictionally engage the example second rear connectors 2600 in the final, assembled position to restrict relative motion therebetween. In some examples, the example first rear connectors 2500 interlock with the example second rear connectors 2600 in the final, assembled position, and optionally include one or more latching elements. In some examples, one or more of the example first rear connectors 2500 include one of a male or female connection member and one or more of the example second rear connectors 2600 include another one of a male or female connection member so that the example first rear connectors 2500 lock into the example second rear connectors 2600 in the final, assembled position.

The example first rear connectors 2500 in FIGS. 17A-17B are shown to be substantially planar, omitting the example second portion 1504 of the example first rear connectors 1500. In some examples, one or more of the example first rear connectors 2500 are not as illustrated, but are rather a conventional male or female connection member and one or more of the example second rear connectors 2600 are not as illustrated, but are another one of a male or female connection member so that such first rear connector locks into the second rear connector 2600 in the final, assembled position. The example first rear connectors 2500 and the example second rear connectors 2600 may comprise any complemen-

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tary shape adapted for engagement such as, but not limited to, complementary right prisms (e.g., wedge) leading to a plateau (e.g., a rectangular shape or square shape having a height equal to that of a base of the right prism), or a rack and pinion arrangement, to name but two examples.

While a certain number of and arrangement of example first rear connectors 2500 and example second rear connectors 2600 are shown in the example license plate holder 2100 of FIGS. 16-19, the present concepts include other numbers of and/or arrangement of the example first rear connectors 2500 and example second rear connectors 2600. For instance, the present concepts include one, two, three or more sets of a first rear connector 2500 and a second rear connector 2600. The one, two, three or more sets of the first rear connector 2500 and the second rear connector 2600 may be arranged asymmetrically (e.g., a different number of and/or position sets of one or more first rear connectors and one or more second rear connectors as between a top and bottom of the example license plate holder, etc.) or on different surfaces of the example base 2200 and example frame 2140.

In some examples, the example first rear connectors 2500 and the example second rear connectors 2600 are positioned on vertical or lateral surfaces of the example frame 2140 and the example base 2200 (e.g., left surfaces and right surfaces in the orientation shown) to facilitate mounting of the frame 2140 to the base 2200 by positioning the frame 2140 vertically off-center relative to the base 2200 during assembly and moving the frame 2140 relative to the base 2200 to center the frame 2140 relative to the base 2200 in the final assembled position. In this configuration, during assembly, the second rear connectors 2600 on the frame 2140 move vertically relative to the first rear connectors 2500 on the base 2200 as the frame 2140 is moved from the initial mounting position to a final assembled position. In the final assembled position, the first rear connectors 2500 and the second rear connectors 2600 engage one another.

Further, while the example third embodiment of the example license plate holder 2000 shown in FIGS. 16-19 shows the example frame 2140 and the example base 2200 being connected at the bottom via two mechanical fasteners (e.g., screws) disposed through the example openings 2112 (see, e.g., FIG. 18), other variants of the design of FIGS. 16-19 optionally reduce the number of mechanical fasteners (e.g., only one screw, only one pin, etc.) or omit the use of mechanical fasteners in favor of complementary mechanical connectors (e.g., clasps, snap-fit connectors, male/female connectors, etc.) formed in the example frame 2140 and the example base 2200.

Similar to the example of FIGS. 8-15B, for example, the example transmissive substrate 1120 may include, for instance, but is not limited to, a substrate of glass, treated or strengthened glass (e.g., Gorilla Glass (Corning), Dragontrail (AGC Inc.), Xensation (Schott AG)), Plexiglass (e.g., Lucite, Acrylite, Perspex, Oroglass Optix, Altuglass, etc.), polycarbonate, acrylic (e.g., injection molded acrylic or polymethyl methacrylate, polystyrene or PMMA. In some examples, the example transmissive substrate 1120 is a uniform substrate of a selected material. In some examples, the example transmissive substrate 1120 defines a plurality of different regions, such as an example first region 1126 (e.g., a central region) having a first set of characteristics (e.g., optical properties, mechanical properties, etc.) and an example second region 1125 (e.g., a peripheral region) having a second set of characteristics. In some examples, two or more regions may have the same characteristics (e.g.,

the example first region **1126** and the example second region **1125** both being having the same optical properties).

In the example of FIGS. **16-19**, the example transmissive substrate **1120** comprises a substrate of glass that is about 0.12 inches thick. In other examples, the example transmissive substrate **1120** comprises a substrate of glass that is less than 0.12 inches thick or greater than 0.12 inches thick, such as between 0.12-0.25 inches thick.

The example frame **2140** of FIG. **16** defines an example central opening (not numbered) corresponding in shape and size to the example first region **1126** of the example transmissive substrate **1120**. The example central opening of the example frame **2140** permits display of indicia, such as the license plate number, on an example first license plate display region **1162**. In the example of FIG. **16**, the example central opening of the example frame **2140** (and the corresponding example first region **1126** of the example transmissive substrate **1120**) further reveal the example second license plate display region **1164**, the example third license plate display region **1161A**, and the example fourth license plate display region **1161B**. As with prior examples, the example central opening of the example frame **2140** and the example first region **1126** of the example transmissive substrate **1120** may alternatively be sized differently (e.g., larger, smaller, etc.) to comport with sizing and/or location requirements of any particular jurisdiction and/or to selectively reveal or obscure, for instance, the example third license plate display region **1161A** and/or the example fourth license plate display region **1161B**.

In the example of FIGS. **16-19**, the example transmissive substrate **1120** defines a plurality of example salients **1122** extending inwardly toward the example first region **1126**. In some examples, these example salients **1122** are sized and spaced so as to partially or fully visually obscure one or more of (e.g., some of or all of) the example mechanical fasteners **1150** used to secure the example license plate **1160** to the example base **2200** and/or vehicle.

In some examples, the example frame **2140** further defines one or more example frame display regions **1146** adjacent the example central opening. The one or more example frame display regions **1146** include text and/or graphics. In some examples, one or more of the example frame display region(s) **1146** include(s) user-selectable text or graphics (e.g., selected from a list of options provided on a website, custom printed, etc.), such as may be associated with sports team logos, organizations, colleges or universities, companies, phrases, causes, locations, hobbies, etc. In some examples, the example transmissive substrate **1120**, rather than the example frame **2140**, defines one or more example frame display regions **1146** (e.g., top, bottom, left and/or right, in any combination) adjacent the example central region (i.e., adjacent the example first region **1126**). The one or more example frame display regions **1146** include text and/or graphics. In some examples, one or more of the example frame display region(s) **1146** include(s) user-selectable text or graphics (e.g., selected from a list of options provided on a website or portal, custom printed, etc.), such as may be associated with sports teams, organizations, colleges or universities, companies, phrases, causes, charities, causes, etc. In another example, the one or more example frame display regions **1146** are omitted and, instead, one or more example transmissive substrate **1120** display regions (not shown) are provided with text and/or graphics (e.g., printed on a front side and/or a back side of the transmissive substrate **1120**, an appliqué on a front side and/or a back side of the transmissive substrate **1120**, etc.). In some examples, such text and/or graphics provided on

one or more regions of the example transmissive substrate **1120** includes user-selectable text or graphics.

The example frame **2140** of FIG. **16** and FIG. **18** defines a plurality of contiguous walls including an example first wall **2141**, an example second wall **2145** (see FIGS. **18-19**), an example third wall **2148** (see FIG. **18**) and an example fourth wall **2149**. The example walls extend rearwardly to define a volume (not numbered) at a rear of the example frame **2140** to receive the example license plate **1160** and at least a portion of the example base **2200**. In some examples, the example walls also extend forwardly to define a volume (not numbered) at the front of the example frame **2140** to receive the example transmissive substrate **1120**. While a plurality of example walls are shown in the example frame **2140** of FIGS. **16** and **18-19**, a lesser number of walls could be provided and/or the configuration of the walls altered. For instance, while not a presently preferred example, the frame **2140** could include a subset of the walls of FIGS. **16** and **18-19**, such as only wall **2149** or only wall **2148**. In such example, the example frame **2140** can connect to the example base **2200** via the subset of the walls (e.g., only wall **2149**, only wall **2148**, etc.) via one of more sets of mechanical connectors and/or mechanical fasteners.

FIGS. **17A-17C** are, respectively, an example front perspective view of an example base **2200** usable in the example license plate holder **2000** of FIG. **16**, an example rear perspective view of the example base of FIG. **17A**, and an example side view of the example base of FIGS. **17A-17B**.

The example base **2200** is shown to include example first rear connectors **2500** at an example upper end and at an example lower end. These example first rear connectors **2500** are configured and positioned on the example base **2200** to matingly engage (e.g., interlocking, frictional engagement, abutting, etc.) example second rear connectors **2600** on the example frame **2140**. In some examples, the example first rear connectors **2500** and the example second rear connectors **2600** are configured to cooperatively resist lateral translation and rotation of the example frame **2140** relative to the example base **2200** in an assembled state. FIGS. **17A-17C** show an example first boss **2615** and an example second boss **2616**, each of the example first boss **2615** and the example second boss **2616** being configured to receive a mechanical fastener (e.g., screws, etc.) to secure the example frame **2140** to the example base **2200**. In some examples, only one example boss is provided (e.g., centrally positioned) and only one mechanical fastener is used to secure the example frame **2140** to the example base **2200**. In some examples the example mechanical fastener(s) are optionally omitted in favor of a more positive lockup between one or more of the example first rear connectors **2500** and a corresponding one or more of the example second rear connectors **2600** (e.g., a locking male/female connector, etc.). In other examples, the example mechanical fastener **1210** could comprise a pin or dowel disposed through the example frame **2140** and into the example base **2200** to prevent relative motion of the example frame **2140** relative to the example base and/or the example license plate **1160**, such as a pin inserted through an upper portion of the example frame **2140** and into the example base, such pin being retained in place by gravity and/or an adhesive.

FIG. **17A** shows an example front **2203** of the example base **2200** and further shows an example central section **2202** including one or more example first rear connectors **2500** (three as shown) at peripheral portions of the central section **2202**, and one or more example first connectors **2205** (e.g., through holes, bosses, male connectors, female con-

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nectors, etc.) within the central section **2202** for attaching the example base **2200** to a vehicle. The example central section **2202** optionally defines a recessed volume within which the example license plate **1160** may be received or a raised volume against which the example license plate **1160** may be secured. In some examples, the example central section **2202** is substantially planar.

FIGS. 17B-17C shows different example perspectives of the example base **2200**, showing the example front **2203** and the example rear **2204** of the example central section **2202**, the one or more example first connectors **2205**, and the example first rear connectors **2500**.

FIG. 18 is a rear perspective view of an example frame **2140** for the example license plate holder **2000** of FIGS. 16-17C. At an example upper end and at an example lower end of the example frame **2140** are provided example second rear connectors **2600** configured and positioned to engage corresponding example first rear connectors **2500** on the example base **2200** to resist lateral translation and rotation of the example frame **2140** relative to the example base **2200**. At an example lower end of the example frame **2140** are provided openings **2112** to receive an example mechanical fastener.

In the example of FIGS. 16-19, the example frame **2140** defines a plurality of example salients **2122** extending inwardly toward the example opening **2126**. In some examples, these example salients **2122** are sized and spaced so as to partially or fully visually obscure one or more of (e.g., some of or all of) the example mechanical fasteners **1150** used to secure the example license plate **1160** to the example base **2200** and/or vehicle.

FIG. 19 is a rear perspective view of the example license plate holder **2000** of FIGS. 16-18 in an assembled state. In the view of FIG. 19, the example mechanical fasteners **1210** (e.g., screws, etc.) shown in FIGS. 9A and 11 are not shown, but are disposed through the example openings **2112** in the example frame **2140** and within the example first second boss **2616** and the example second boss (not numbered in FIG. 19) formed in the example base **2200** to secure the example frame **2140** to the example base **2200**. In some examples, only one example mechanical fastener is used to secure the example frame **2140** to the example base **2200**. In some examples the example mechanical fastener(s) **1210** are optionally omitted in favor of a more positive lockup between one or more of the example first rear connectors **2500** and a corresponding one or more of the example second rear connectors **2600** (e.g., a locking male/female connector, etc.). In other examples, the example mechanical fastener **1210** could comprise a pin or dowel disposed to prevent relative motion of the example frame **2140** relative to the example base and/or the example license plate **2160**, such as a pin inserted through an upper portion of the example frame **2140** and into the example base, such pin being retained in place by gravity and/or an adhesive (e.g., Loctite, etc.).

In any of the preceding examples and variants thereof, the license plate holder may further include an electrical connector to electrically connect to a license plate lighting circuit of a vehicle and at least one lighting element to illuminate the central opening (**1142**, **2126**) and the at least one display region (e.g., the frame display region **1146**, the transmissive substrate display region **1146**, etc.).

Although certain example apparatus, articles of manufacture and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. To the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of

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this patent. For example, in one instantiation of the present concepts, the transmissive substrate is formed as a unitary piece to combine the structural elements of the aforementioned example frames (e.g., **140**, **1140**, etc.) to avoid the need for a separate frame, with the transmissive substrate being connected directly to the license plate (e.g., **160**, **1160**, etc.) and/or the base (e.g., **200**, **1200**, etc.).

What is claimed is:

1. A license plate holder, comprising:

a frame defining a central opening to permit display of indicia on at least one license plate display region of a license plate and further defining at least one frame display region adjacent the central opening; and

a transmissive substrate comprising a plurality of salients disposed and sized to overlay one or more connectors attaching a license plate base to a vehicle to partially or fully obscure the one or more connectors attaching the license plate base to the vehicle;

wherein, in an assembled state with the transmissive substrate removably connected to the frame and the frame interposed between the transmissive substrate and the license plate base, a central region of the transmissive substrate is positioned in front of the central opening of the frame and the at least one license plate display region and a peripheral region of the transmissive substrate is positioned in front of the at least one frame display region of the frame.

2. The license plate holder of claim 1, wherein the frame comprises one or more front connectors to removably connect the frame to the transmissive substrate.

3. The license plate holder of claim 2, wherein the one or more front connectors comprise one or more snap-fit connectors.

4. The license plate holder of claim 1, wherein the frame comprises one or more rear connectors to removably connect the frame to the license plate base.

5. The license plate holder of claim 4, wherein the one or more rear connectors comprise one or more snap-fit connectors.

6. The license plate holder of claim 4, wherein the one or more rear connectors comprise one or more fixed connectors at a rear portion of the frame, the one or more fixed connectors extending in a direction from a first lateral edge of the frame toward a second lateral edge of the frame.

7. The license plate holder of claim 4, wherein the one or more rear connectors comprise a plurality of fixed connectors spaced apart along a rear portion of the frame, each of the plurality of fixed connectors extending in a direction from a first lateral edge of the frame toward a second lateral edge of the frame.

8. The license plate holder of claim 7, wherein the plurality of fixed connectors comprises a first plurality of fixed connectors spaced apart along an upper rear portion of the frame.

9. The license plate holder of claim 7, wherein the plurality of fixed connectors comprises a second plurality of fixed connectors spaced apart along a lower rear portion of the frame.

10. The license plate holder of claim 7, wherein the plurality of fixed connectors comprises a first plurality of fixed connectors spaced apart along an upper rear portion of the frame and a second plurality of fixed connectors spaced apart along a lower rear portion of the frame.

11. The license plate holder of claim 1, wherein the frame comprises one or more front connectors to removably con-

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nect the frame to the transmissive substrate and comprises one or more rear connectors to removably connect the frame to the license plate base.

12. The license plate holder of claim 1, wherein a first region of the transmissive substrate comprises a first characteristic and a second region of the transmissive substrate comprises a second characteristic different than the first characteristic.

13. The license plate holder of claim 12, wherein the first region and the first characteristic corresponds to the central region of the transmissive substrate.

14. The license plate holder of claim 13, wherein the second region and the second characteristic corresponds to the peripheral region of the transmissive substrate.

15. The license plate holder of claim 12, wherein the first characteristic comprises a first optical property and the second characteristic comprises a second optical property, or wherein the first characteristic comprises a first surface finish and the second characteristic comprises a second surface finish.

16. The license plate holder of claim 1,

wherein the at least one frame display region comprises a plurality of frame display regions, comprising a first frame display region provided at one of a first lateral portion of the frame, a second lateral portion of the frame, an upper portion of the frame, or a lower portion of the frame, and a second frame display region provided at another one of the first lateral portion of the frame, the second lateral portion of the frame, the upper portion of the frame, or the lower portion of the frame, and

wherein each of the plurality of frame display regions includes text, graphics, or text and graphics printed on, applied to, or integrated into each of the plurality of frame display regions.

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17. The license plate holder of claim 1, wherein the frame comprises a plurality of frame display regions disposed and sized to underlie the plurality of salients of the transmissive substrate in an assembled state, and

wherein the at least one frame display region includes text, graphics, or text and graphics printed on, applied to, or integrated into the at least one frame display region.

18. The license plate holder of claim 1, wherein a first region of the transmissive substrate, corresponding to the central region of the transmissive substrate, comprises a first characteristic, wherein a second region of the transmissive substrate, corresponding to the plurality of salients, comprises a second characteristic different than the first characteristic, and

wherein the first characteristic comprises a first optical property and the second characteristic comprises a second optical property.

19. A license plate holder, comprising:

a frame defining a central opening to permit display of indicia on at least one license plate display region of a license plate and further defining at least one frame display region adjacent the central opening; and

a transmissive substrate; wherein

in an assembled state with the frame interposed between the transmissive substrate and a license plate base, a central region of the transmissive substrate is positioned in front of the central opening of the frame and the at least one license plate display region and a peripheral region of the transmissive substrate is positioned in front of the at least one frame display region of the frame;

the frame comprises one or more front connectors to removably connect the frame to transmissive substrate; and

the frame comprises one or more rear connectors to removably connect the frame to the license plate base.

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